

Electricity and Phenomena of Bioelectricity

General Introduction to Bioelectricity

The human body is not merely a collection of organs and tissues: it is also a living electrical system. At every moment, ionic currents circulate, potential differences appear, and electrical signals are produced, transmitted, and interpreted. These phenomena constitute what is known as bioelectricity.

Bioelectricity studies the origin and effects of electrical phenomena in biological systems, particularly in excitable cells such as neurons, muscle fibers, and cardiac cells. It is based on simple physical laws—derived from electricity, electrostatics, and electro kinetics—applied to a complex biological environment.

Vital functions such as muscle contraction, nerve transmission, heartbeat, and brain activity directly depend on electrical potential differences and ion movements across cell membranes. Without these electrical phenomena, coordination, communication, and physiological regulation would be impossible.

For medical students, bioelectricity is a fundamental tool for understanding: it links physical principles to clinical applications such as electrocardiograms (ECG), electroencephalograms (EEG), nerve stimulation, and certain therapeutic techniques using electrical currents.

This chapter aims to provide essential concepts of bioelectricity, emphasizing key physical concepts, their biological interpretation, and their medical importance, without excessive mathematical formalism.

1. Origin of the word electricity: Historically, the word electricity comes from the Greek *ēlektron*, which means amber. Since antiquity, the Greeks observed that rubbed amber attracted small light objects. In the 17th century, William Gilbert introduced the scientific term *electricus*, marking the birth of electricity as a physical concept.

2. Producing Electricity by Friction: When two different materials are rubbed together (for example, a balloon and hair, or a plastic ruler and a wool sweater), there is a transfer of electrons from one material to the other.

The material that gains electrons acquires a negative charge, while the one that loses electrons acquires a positive charge.

This phenomenon is called electrification by friction and is one of the simplest forms of static electricity production.

3. Electricity in the Human Body: A Communication Network: The human body is a vast and complex electrical circuit. Electricity is at the core of its communication system, enabling the transmission of vital information at incredible speed.

3.A. The Nervous System: The neurons, the fundamental cells of our brain and spinal cord, generate electrical impulses called action potentials. These signals travel along nerve fibers to transmit messages from the brain to the muscles, organs, and all parts of the body. This internal electrical network allows us to think, feel, move, and react to our environment in a fraction of a second.

3.B. The Heart: Each beat of our heart results from an electrical impulse. A group of specialized cells, the sinoatrial node, generates an electrical signal that propagates through the cardiac muscle, causing rhythmic contraction. It is this electrical activity that is measured by an electrocardiogram (ECG) to diagnose heart abnormalities.

3.C. Muscle Contraction: The ability of our muscles to contract and generate movement is also of electrical origin. When electrical signals (nerve impulses) reach muscle cells, they trigger the opening of ion channels, allowing ions (such as calcium) to circulate. This flow of ions is the driving force that causes muscle fibers to contract.

4. Birth of bioelectricity: Understanding electricity in living organisms is not recent and has paved the way for a new branch of science. The most famous example is that of Luigi Galvani. In the 18th century, this Italian anatomist conducted a series of experiments on frogs that revolutionized biology. In 1780, he observed that the muscle of a dead frog's leg contracted when in contact with two different metals. Galvani interpreted this phenomenon as a manifestation of "animal electricity." These works laid the foundation for the study of electrical currents in living tissues and inspired research on the mechanisms of nerve impulses and muscle contraction.

5. Development of electrophysiology: Although his rival Alessandro Volta demonstrated that the current originated from metals and not animals, Galvani established the foundations of electrophysiology, a science that studies electrical phenomena in living organisms. His work paved the way for future research on electric fish (such as electric eels) and on the role of ionic currents in cells, leading to modern medical technologies such as pacemakers and electroencephalograms (EEG).

6. The Galvani – Volta controversy: Galvani's experiment sparked a major scientific debate with Alessandro Volta.

- Galvani defended the idea of "animal electricity" produced by living tissues.
- Volta, on the other hand, believed that electricity came from contact between two metals connected by a moist conductor.

👉 This controversy was extraordinarily fruitful : it led to the birth of bioelectricity and the invention of the battery, opening two major pathways of modern science.

In other words: **Galvani** paved the way for **neurophysiology** (the electrical role of nerves and muscles). **Volta** laid the foundation of **electrochemistry** and the battery.

II. Fundamental Electrical Quantities:

1. Electric Charge is an **intrinsic property** of matter, like mass. It is responsible for all electromagnetic interactions and phenomena.

- Symbol: q (or sometimes Q for a total charge).
- Unit: Coulomb (C) in the International System (SI).
- Two types of charges: Electric charge manifests in two forms:
 - **Positive charge**: Carried by **protons** in the nucleus of an atom.
 - **Negative charge**: Carried by **electrons** that orbit around the nucleus.

Electrical neutrality: An atom (or an object) is said to be **neutral** when it contains an equal number of protons and electrons. Positive and negative charges cancel each other out.

- An object becomes **negatively charged** if it **gains** electrons (excess of electrons).
- An object becomes **positively charged** if it **loses** electrons (deficit of electrons).
- Living matter contains many ions (Na^+ , K^+ , Cl^- , Ca^{2+}).
- Fundamental law: charges of the same sign repel each other, opposite signs attract each other.

2. Electric Current

- ✚ Definition: orderly movement of charges.
- ✚ Symbol : I .
- ✚ Formula : $I = \Delta Q / \Delta t$.
- ✚ Unit : Ampere (A).
- ✚ In a neuron, the current is due to the passage of ions through the membrane.

Example: If 10^{-9} C (nanocoulomb) passes into 1 ms, then: $I = 10^{-9} / 10^{-3} = 10^{-6}$ A = 1 μA .

3. Electric potential difference (voltage)

- Symbol: U or V .
- Definition: work needed to move a charge between two points: $U = W/q$.
- Unit: Volt (V).

✚ **Biological example** : The resting potential of a nerve cell is approximately -70 mV.

4. Electrical Resistance

- Symbol: R .
- Ohm's Law: $U = R \times I$.
- Unit: Ohm (Ω).

✚ **In biology** : the cell membrane acts as a resistance to the passage of ions (it limits their flow).

5. Electrical Power

- Symbol: P .
- Formula: $P = U \times I$.
- Unit: Watt (W).

👉 **Biological Example:** A neuron consuming $I = 1 \text{ nA}$ at $U = 70 \text{ mV}$:

$$P = (70 \times 10^{-3}) (10^{-9}) = 7 \times 10^{-11} \text{ W}.$$

This demonstrates that cellular functioning consumes minimal but constant energy.

III. Useful Mathematical Reminders

Common orders of magnitude in bioelectricity:

- Voltage: millivolt (mV)
- Current: nanoampere (nA)
- Resistance: kilohm ($k\Omega$) to megaohm ($M\Omega$).

Application Exercises:

1. Calculate the current corresponding to the passage of $5 \times 10^{-6} \text{ C}$ in 2 ms.

Solution: $I = Q/\Delta t = (5 \times 10^{-6}) / (2 \times 10^{-3}) = 2.5 \times 10^{-3} \text{ A} = 2.5 \text{ mA}$

2. What energy is needed to move a charge of 2 C under a voltage of 12 V?

Solution: $W = Q \times U = 2 \times 12 = 24 \text{ J}$

3. A neuron consumes 1 nA under a voltage of 70 mV. Calculate the power.

Solution: $P = U \times I = (70 \times 10^{-3}) \times (1 \times 10^{-9}) = 70 \times 10^{-12} \text{ W} = 70 \text{ pW}.$

4. A current of 2 nA passing through a resistance of $10^6 \Omega \rightarrow U = R \times I = 2 \text{ mV}.$

⚡ Electrostatics

I – Introduction: Electrostatics is the branch of electricity that studies stationary electric charges, the forces they exert on each other, and the electric fields they create in space. It focuses on the laws governing the interaction between charges, such as Coulomb's law, and provides the necessary foundations to understand electrical phenomena and their applications, both in fundamental physics and in biology or technology.

II - ⚡ I. Electric Force

1. Coulomb's Law: Two point electric charges exert on each other a **force** at a distance proportional to the product of their charges and inversely proportional to the square of the distance separating them. This can be mathematically expressed as:

$$F = k_e \frac{(q_1 * q_2)}{r^2}$$

- F : electrostatic force (N, Newton)
- q_1, q_2 electric charges (C, Coulomb)
- r : distance between the two charges (m)
- k_e : Coulomb's constant

👉 Interpretation :

- Charges of the same sign $\rightarrow$ repulsive force
 - Charges of opposite signs $\rightarrow$ attractive force
- Tapez une équation ici.

Coulomb's Constant : often denoted as **k**, is a fundamental constant; its value is

$$k = 8.987\,551\,792\,3 \times 10^9 \text{ N}\cdot\text{m}^2\cdot\text{C}^{-2}$$

In general, this value is rounded for calculations to: $k \approx 9 \times 10^9 \text{ N}\cdot\text{m}^2\cdot\text{C}^{-2}$

$$\text{👉 } k_e = \frac{1}{4\pi\epsilon}$$

Permittivity ϵ : A physical quantity that indicates a medium's ability to allow an electric field to pass through while partially opposing it.

- ✚ Permittivity of free space: Universal constant that determines the strength of electrostatic interaction in a vacuum. Its value is : 8.854×10^{-12} . It expresses itself in

the system (MKSA) in $\text{A}^2 \cdot \text{s}^4 \cdot \text{kg}^{-1} \cdot \text{m}^{-3}$ 👉 or $\frac{\text{C}^2}{\text{N} \cdot \text{m}^2}$

- ✚ Relative permittivity or dielectric constant: A dimensionless factor that indicates how much the electric field is reduced within a material compared to a vacuum.

- ✚ Absolute permittivity of the medium:

$$\text{👉 } \epsilon = \epsilon_r \epsilon_0$$

2. Principle of superposition of forces: In electrostatics, a charge can be influenced by several other charges simultaneously. In this case, **the total force is simply the vector sum of the forces exerted by each charge individually**. This is known as the **principle of superposition**.

Formally, if a charge q_0 is placed in a system of n charges ($q_1, q_2, \dots, q_n$), then:

$$\text{👉 } \vec{F}_{tot} = \sum_{i=0}^n \vec{F}_i$$

Each force $\vec{F}_i$, from $i=0$ to $i=n$, is calculated using the **Coulomb's law**, then the vectors are added considering their direction and sense.

✚ **Simple example :** Three aligned charges : $q_1 = +2 \mu\text{C}$ on the left, $q_2 = -3 \mu\text{C}$ to the right, and $q_0 = +1 \mu\text{C}$ in the center. q_0 subjected to a repulsive force coming from q_1 (to the right) and an attractive force coming from q_2 (also towards the right). The two forces add up, so the total force is greater than if only one charge was acting.

III. The Electric Field

Imagine the space around an electric charge. If you place another charge nearby, it experiences an electric force. The electric field is the mathematical and physical description of this influence space. It is a property of space itself, created by a source charge, which allows the determination of the force that would act on any other charge placed at a given point.

1. Mathematical Definition and Formula

The electric field $\vec{E}$ at a point is defined as the **electric force** $\vec{F}$ subjected to a **test load** q_0 placed at this point, divided by the value of this test load.


$$\vec{E} = \frac{\vec{F}}{q_0}$$

- $\vec{E}$ Electric field vector (in Newtons per Coulomb, N/C, or more commonly in Volts per meter, V/m).
- $\vec{F}$: Electric force vector (in Newtons, N).
- q_0 : Charge of test (in Coulombs, C).

Key Points :

- ✚ The **electric field** is a **vector field**: at each point in space, it has an intensity (numeric value) and a direction.
- ✚ The trial charge q_0 is supposed to be small enough not to disturb the field created by the source charges.
- ✚ The direction of the field $\vec{E}$ Electric field is defined as the direction of the force that would act on a **positive test charge**.

2. The Electric Field of a Point Charge

For a point charge Q (la source), Coulomb's law allows us to find a direct expression of the field it creates at a distance r .

The force on a charge q_0 is: $F = k \frac{|Q||q_0|}{r^2}$

Using the definition $\vec{E} = \frac{\vec{F}}{q_0}$, one obtains the field intensity: $E = \frac{F}{q_0} = k \frac{|Q|}{r^2}$

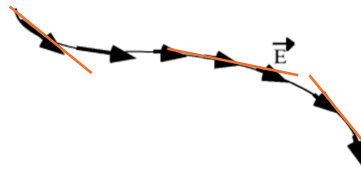
Where is k Coulomb's constant.

Direction of the field :

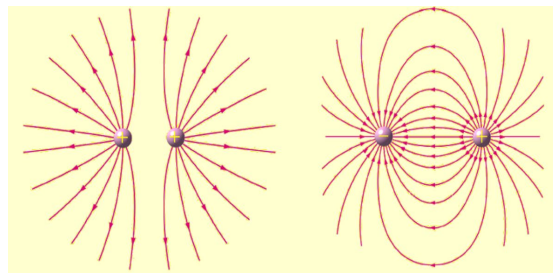
- If the source load Q is **positive**, the field is **outgoing** (it pushes a positive test load).
- If the source load Q negative, the field is **inward** (it attracts a positive test charge).

For a set of source charges, the total electric field at a point is the **vector sum** of the individual fields created by each charge (superposition principle).

3. Electric Field Lines: A line of field is called: a curve that, at each of its points, is tangent to the electric field vector $\vec{E}$.



- **Direction:** The tangent to a field line at a point gives the direction of the vector $\vec{E}$ at this point.
- **Sense:** The field lines always originate from **positive charges** and go towards **negative charges**.
- **Intensity:** The closer the lines are **together** (dense), the stronger the field. Where the lines are spaced apart, the field is weak.
- **Intersection:** The field lines never cross because the electric field at a point has a unique direction.



4. Uniform Electric Field: An electric field is uniform in a region of space if the vector field is the same at all points within that region, (same direction, same sense, same magnitude). When the field is uniform, the field lines are straight and parallel.

5. Superposition of Electric Fields: When multiple electric charges are present, the total electric field at a point **M** It is the vector sum of the individual fields created by each charge, as if it were alone.

👉 Mathematically:

$$\vec{E}_{total}(M) = \vec{E}_1(M) + \vec{E}_2(M) + \vec{E}_3(M) + \dots + \vec{E}_n(M)$$

📁 III. Electric Potential

🧩 **Definition:** The electric potential energy, denoted E_p or U , It is a scalar quantity that represents the work required to bring one or more electric charges from infinity to their current positions in an electrostatic field.

It reflects the energy stored in the charge system due to their relative positions and can be released (often in the form of kinetic energy) when the charges are allowed to move.

🏆 Fundamental formula (case of two point charges)

$$U = k \frac{Qq}{r}$$

With :

- ✚ U : Electrical potential energy (in joules, J)
- ✚ $k = \frac{1}{4\pi\epsilon_0}$: Coulomb's constant
- ✚ Q, q : values of charges (in coulombs, C)
- ✚ r : distance between the two charges (in meters, m)

💬 Important remarks

- If Q and q have the same signe $\rightarrow U > 0 \rightarrow$ energy must be **provided** to bring them closer together.
- If Q and q have opposite signes $\rightarrow U < 0 \rightarrow$ the system **releases** energy as the charges move closer together.

⚡ Potential energy for multiple charges

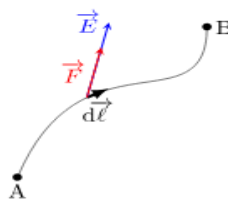
If a charge q_0 is placed in a system of multiple charges $Q_1, Q_2, \dots, Q_n$, therefore:

$$U = kq_0 \sum_{i=1}^n \frac{Q_i}{r_i}$$

Where r_i is the distance between q_0 and Q_i .

It is the **scalar sum**, not the vector sum, because energy is a **scalar quantity**.

⚡ **Relationship between the electrostatic potential and the electric field** : suppose a charge q moves from A to B within a given electric field $\vec{E}$.



The change in potential energy of the charge:

$$E_{pA} - E_{pB} = q(V_A - V_B) = W_{A \rightarrow B}$$

$W_{A \rightarrow B}$ is the work done on the charge as it moves from A to B : Since the electric force acting on the charge q , from which we have $(\vec{F} = q\vec{E})$ Therefore :

$$W_{A \rightarrow B} = \int_A^B \vec{F} \cdot d\vec{l} = \int_A^B q\vec{E} \cdot d\vec{l} = q(V_A - V_B)$$

On q , we will obtain the relationship between the electric field and the potential difference :

$$V_A - V_B = \int_A^B \vec{E} \cdot d\vec{l} = -(V_B - V_A)$$

If A and B coincide (the integration path is a closed curve), therefore:

$$\int_A^B \vec{E} \cdot d\vec{l} = 0 ; E_l: \text{component of } \vec{E} \text{ along the trajectory}$$

$$E_l dl = -dV \quad \text{or} \quad E_l = -\frac{\delta V}{\delta l}$$

The electric field $\vec{E}$ is related to the variation of the electric potential:

$$\vec{E} = -\vec{\nabla}V = -\overrightarrow{grad}V$$

This indicates that the electric field vector E is therefore a vector directed from high potentials toward lower potentials.

Where, $\vec{\nabla} = \overrightarrow{grad}$

1. In Cartesian coordinates (x, y, z)

$$\vec{E} = -\left(\frac{\partial V}{\partial x} \vec{i} + \frac{\partial V}{\partial y} \vec{j} + \frac{\partial V}{\partial z} \vec{k}\right)$$

Or component by component: $E_x = -\frac{\partial V}{\partial x}, E_y = -\frac{\partial V}{\partial y}, E_z = -\frac{\partial V}{\partial z}$

♦ *Typical case: uniform field between two plates* → Variation along a single direction (often x).

2. In cylindrical coordinates (r, θ, z): $\vec{E} = -\left(\frac{\partial V}{\partial r} \vec{e}_r + \frac{1}{r} \frac{\partial V}{\partial \theta} \vec{e}_\theta + \frac{\partial V}{\partial z} \vec{e}_z\right)$

where:

$$E_r = -\frac{\partial V}{\partial r}, E_\theta = -\frac{1}{r} \frac{\partial V}{\partial \theta}, E_z = -\frac{\partial V}{\partial z}$$

♦ *Typical case: Field around a **charged wire** (cylindrical symmetry).*

3. In spherical coordinates (r, θ, φ): $\vec{E} = -\left(\frac{\partial V}{\partial r} \vec{e}_r + \frac{1}{r} \frac{\partial V}{\partial \theta} \vec{e}_\theta + \frac{1}{r \sin \theta} \frac{\partial V}{\partial \phi} \vec{e}_\phi\right)$

where :

$$E_r = -\frac{\partial V}{\partial r}, E_\theta = -\frac{1}{r} \frac{\partial V}{\partial \theta}, E_\phi = -\frac{1}{r \sin \theta} \frac{\partial V}{\partial \phi}$$

♦ *Typical case: Field around a **point charge** (spherical symmetry).*

 In the case of a uniform field: $E = -\frac{\Delta V}{\Delta l}$

- E: intensity of the electric field (in $\text{N} \cdot \text{C}^{-1}$ or $\text{V} \cdot \text{m}^{-1}$)
- ΔV : potential difference between two points (in volts, V)
- Δl : distance between these two points (in meters, m)
- The sign « - » indicates that the electric field **points in the direction where the potential decreases**.



Physical interpretation

- **The electric field** indicates the direction and sense in which a positive charge would be accelerated.
- **The electric potential** indicates the "energetic height" of a point within the field.

Thus, the field is analogous to the **slope of a terrain** : the steeper the slope, the more intense the field.



Note:

In a region with a uniform field (for example, between two capacitor plates), **the field lines** are **perpendicular to the equipotential surfaces**, and **the field is constant** between them.



Electric potential of a point charge



Definition: The **electric potential** $V(M)$ at point M is equal to the **work done by the electric field** (or supplied by an opposing external force) to **bring a unit positive charge from infinity to point M , without acceleration**.

In other words: the potential measures the capacity of the electric field to perform work on a test charge.

Mathematically, this definition is expressed by an integral along a path (which, thanks to the conservative nature of the electrostatic field, does not depend on the path chosen) :

$$V(M) = - \int_{\infty}^M \vec{E} \cdot d\vec{l}$$

♦ **Case of a single point charge:** If the electrical field E is **created by a point charge q** placed at a point P , so the **electric potential** at a point M , situated at a distance $r = PM$, is given by:

$$V(M) = k \frac{q}{r} = \frac{1}{4\pi\epsilon_0} \times \frac{q}{r}$$

♦ **Case of a set of point charges:** If the electric field is created by **several point charges q_i** located at points P_i , then the total potential at point M is the algebraic sum of the potentials produced by each charge:

$$V(M) = k \sum_{i=1}^N \frac{q_i}{r_i}$$

Where r_i is the distance between the charge q_i and the point M .



Notes:

- The electric field is a **scalar quantity** (unlike the field $\vec{E}$, which is vectorial).
- By convention, the **potential at infinity is zero**: $V(\infty) = 0$.
- If the charges are **positive**; $V(M) > 0$, if they are **negative**; $V(M) < 0$.

V – Flux of the electric field vector

✚ 1. Definition

The flux of the electric field through a given surface measures the **quantity of field lines** passing through that surface.

In other words, it indicates the **intensity of the electric field “passing” through a surface**.

The flux is denoted by Φ (Greek letter “phi”).

$$\phi = \iint \vec{E} \cdot \vec{dS}$$

avec :

- Φ : flux du champ électrique (in **volt.meter**, $V \cdot m$ or $N \cdot m^2 \cdot C^{-1}$),
- $\vec{E}$: electric field vector,
- $\vec{dS}$: elementary surface vector, oriented perpendicularly to the surface,
- $\vec{E} \cdot \vec{dS} = E dS \cos \theta$, where θ is the angle between the field and the normal to the surface.



2. Case of a uniform field: if the field $\vec{E}$ is **constant over the entire flat surface** S , the flux becomes:

$$\Phi = E S \cos \theta$$

- If $\theta = 0^\circ$ (field perpendicular to the surface) $\rightarrow$ **maximum** flux
- If $\theta = 90^\circ$ (field parallel to the surface) $\rightarrow$ **zero** flux

◇ 3. Physical interpretation

- The flux is positive if the field lines **exit (leave)** the surface.
- It is **negative** if the field lines **enter** the surface.

Thus, the flux depends on :

1. The **magnitude (value)** of the field E ,
2. The **area of the surface**,
3. The **orientation** of the surface relative to the field.



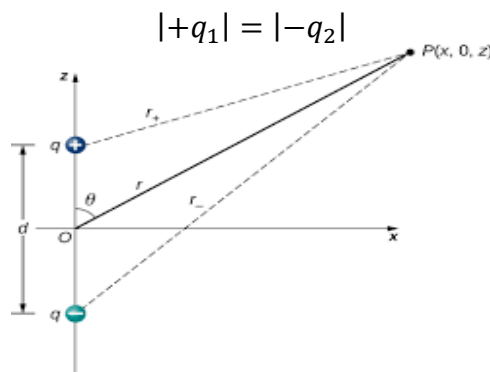
Note: The electric field flux is a **scalar quantity**, even though the field and the surface are vectors.



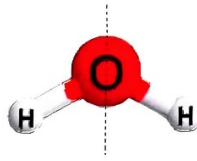
VI – The electric dipole

✚ 1. Definition

An **electric dipole** is a system formed of **two-point charges equal in absolute value and of opposite signs**: $+q$ and $-q$ separated by a distance d that is very small compared to the observation distances.



This model is very common: it describes, for example, **polar molecules** like water or certain electrical devices (antennas).



2. Dipole moment

The dipole moment is a vector characteristic of the dipole:

$$\vec{p} = q \vec{d}$$

- + q : value of the charge (in C),
- + $\vec{d}$: vector going from the negative charge toward the positive charge,
- + $\vec{p}$: dipole moment (in C·m).

◇ VI-1. Potential created in M by the dipole

Let a point situated at a distance r from the center of the dipole, making an angle θ with the dipole axis.

The **electrostatic potential** created by the dipole M is:

$$V(M) = \frac{1}{4\pi\epsilon_0} \frac{\vec{p} \cdot \vec{u}_r}{r^2}$$

- + $\vec{u}_r$: unit vector directed from the center of the dipole toward M ,
- + $\vec{p} \cdot \vec{u}_r = p \cos \theta$.

Ainsi :

$$V(M) = \frac{1}{4\pi\epsilon_0} \frac{p \cos \theta}{r^2}$$

👉 This potential decreases faster than that of a single charge ($1/r^2$ au lieu de $1/r$).

⚙️ **VI-2. Torque of forces exerted on a dipole in an electric field:** When a **dipole** is placed in a **uniform electric field** $\vec{E}$, the two charges undergo:

- + A force $\vec{F}_1 = +q\vec{E}$ on the positive charge,
- + A force $\vec{F}_2 = -q\vec{E}$ on the negative charge.

These two forces are **opposite**, **parallel**, and **non-collinear**, so they form a **couple of forces**. The moment of this torque tends to **align the dipole with the electric field**.

$$\vec{M} = \vec{p} \times \vec{E}$$

If $\vec{p}$ is **aligned with** $\vec{E} \rightarrow$ stable equilibrium.

If $\vec{p}$ is **anti-aligned** $\rightarrow$ unstable equilibrium.

💡 **6. Potential energy of the dipole in an electric field:** the torque's work defines an associated potential energy:

$$U = -\vec{p} \cdot \vec{E} = -pE \cos \theta$$

The energy is minimal when the dipole is **aligned with the field** ($\theta = 0^\circ$).

- It is maximum when it is **opposite** the field ($\theta = 180^\circ$).

VI. Biological Examples and Applications

✚ Cell membranes:

Ionic charges (Na^+ , K^+ , Cl^-) create an electric field through the membrane.

✚ Neurons :

The action potential is a rapid variation of the membrane electric field.

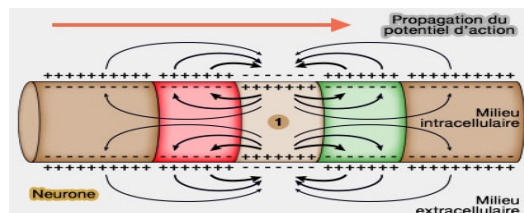
✚ Proteins and DNA:

Their structure depends on internal electrostatic interactions.

🧠 IV. Biological applications of electrostatics:

1. Electricity in the human body: The human body uses **electrical phenomena** to transmit information:

1. Neurons **create** and propagate **action potentials**, thanks to the movements of ions (Na^+ , K^+ , Cl^-) across cell membranes.
1. These motions rely on **electrostatic forces** and **potential differences**.



■ Diagram: Propagation of an action potential along a neuron.

2. Cell membrane and resting potential: Each cell has a **membrane potential** of approximately -70 mV .

This potential comes from:

- The unequal distribution of ions on either side of the membrane,
- Selective ion channel permeability.

Potentiel de membrane = influence directe des forces électrostatiques

3. Molecular dipoles and biological interactions: Many biological molecules (such as water or proteins) have a **dipole moment**.

These dipoles interact electrostatically, explaining:

- The cohesion of the water,
- The structure of proteins,
- DNA binding.

Conclusion:

Electrostatics makes it possible to understand:

- Fundamental **interactions** between charges,
- The **structure of the fields and potentials**,
- And their **essential biological and medical applications** .

She thus connects **physics** to biology, showing that Coulomb's laws and the electric field govern both macroscopic objects and living cells.

1. Conductors, insulators and capacitors

A. Conductor:

 **Definition:** A material that **allows electric charges** (free electrons) to circulate easily.

 **Examples:** Metals such as copper, aluminum, gold, silver.

 **Main properties:**

1. Electrical charges **move freely** indoors.
2. At **electrostatic equilibrium**, the **electric field inside is zero**.
3. The charges **are distributed over the surface** of the conductor.
4. The conductor has **the same potential**.

A.1. The equilibrium of a conductor : In electrostatics, the equilibrium of a conductor is defined by the state in which **the electric charges inside the conductor are no longer moving** ; that is to say, there is no **electric current** flowing within it.

More precisely:

1. **Electric field inside:** The electric field inside a conductor in electrostatic equilibrium is **$\vec{E}$ zero ($\vec{E} = 0$)**.
2. **Surface loads:** Free charges are distributed only **on the outer surface** of the conductor.
3. **Field direction to surface:** The electric field on the surface of the conductor is **perpendicular to the surface**.
4. **Electrical potential:** The entire conductor has **the same electric potential**, so there is no potential difference inside.

In summary, a conductor is in electrostatic equilibrium when **the charges have finished redistributing, no charge is moving inside, and the internal electric field is zero**.

A.2. Properties of a conductor in equilibrium

a. The charge distribution is on the surface: In an electrostatically balanced conductor, the free charges are distributed only **on the surface** of the conductor.

Consider a point inside the conductor.
According to **Gauss's theorem**:

$$\Phi = \iint \vec{E} \cdot \vec{ds} = \sum \frac{q_{\text{int}}}{\epsilon_0} = \iiint \frac{\rho \, dv}{\epsilon_0}$$

However, inside the balanced conductor:

$$\vec{E} = 0 \Rightarrow \rho = 0$$

This means that the **charge density** inside the conductor is **zero**. Positive and negative charges compensate for each other **perfectly**, and any **excess charges are eventually** added to the driver are **distributed only on the outer surface**.

b. The conductor is an equipotential volume : Since the electric field is zero inside:

$$\vec{E} = -\nabla V = \vec{0} \Rightarrow \nabla V = 0 \Rightarrow V = \text{constant}$$

Thus, **the electrical potential is the same at all points of the conductor**. The conductor's volume is therefore **equipotential**.

c. The surface of the conductor is an equipotential: Since the potential is constant inside, it maintains this value across the entire surface. The surface of the conductor is therefore an equipotential surface, and the electric field lines near the surface are perpendicular (normal) to it.

d. Field in the vicinity of a balanced conductor

Coulomb's theorem : The **electrostatic field** created **in a vacuum**, in the : The **electrostatic field** created **in a vacuum**, in the **immediate vicinity** of a conductor in electrostatic equilibrium, has the following properties:

- **Direction** : The field is **normal (perpendicular)** to the surface of the conductor. This means that field lines exit or enter **perpendicular** to the surface, never tangentially.
- **Value (Magnitude)**: The magnitude of the electric field is proportional to the surface charge density, according to the following relationship : $E = \frac{\sigma}{\epsilon_0}$

Where :

- ✚ E : the Magnitude of the Electric Field (in N/C),
- ✚ σ : Surface Charge Density (in C/m²),
- ✚ ϵ_0 : Vacuum permittivity ($8,85 \times 10^{-12}$ F/m).

👉 Notes:

- The field **exists only outside** the conductor, because inside $E = 0$.
- The **direction of the field** depends on the sign of the charges at the surface:
 - If $\sigma > 0$, field lines come out of the conductor.
 - If $\sigma < 0$, they enter it.

e. Capacitance of a conductor at equilibrium

Definition : The **capacitance** of an insulated conductor is the **coefficient of positive proportionality** between the **electric charge** Q that it carries and its **potential** V :

$$C = \frac{Q}{V}$$

Unity: Capacity is expressed in the **International System (SI)** in **farad (F)**.

$$1 \text{ F} = 1 \text{ C/V}$$

👉 **Note:** The capacity does not depend on the load or the potential, but only on the geometric characteristics of the conductor (shape, dimensions) and the nature of the surrounding environment.

Example : for an **isolated conducting sphere** of radius R , we have: $C = 4\pi\epsilon_0 R$

f. Energy of a conductor: Electrostatic **energy** ξ (ou W) stored in a charged conductor at equilibrium is given by:

$$\xi = \frac{1}{2} CV^2 = \frac{1}{2} \frac{Q^2}{C}$$

👉 **Interpretation:** This energy corresponds to the **work necessary** to gradually bring the charge Q on the conductor to its final potential V .

The Power of Points (Experimental Observation) : The **electrostatic field on the surface** of a conductor is not **uniform** :

- It is **more intense in the convex regions**,
- And **weaker in concave regions**.



Consequence:

- The field E is all **the stronger as the curvature is greater** in the vicinity of the points. Charges **tend to build up** on these sharp areas.

This phenomenon is the origin of the **power of points**: the sharp tips of a **conductor promote electrical discharges** (for example, in lightning rods or electrostatic generators).



To remember: An electrostatically balanced conductor:

- has zero electric field inside;
- carries excess loads on its surface;
- is equipotential (same potential everywhere);
- has a normal field at the surface, proportional to the charge density;
- accumulates charges at the points where the field is at its maximum.

👉 B. Insulator (or Dielectric):

- ✚ **Definition:** A material that **prevents or severely restricts the movement of electrical charges**.
- ✚ **Examples:** Glass, dry wood, rubber, plastic.
- ✚ **Main properties:**
 1. Charges **do not move freely** inside.
 2. An electric field may exist **inside** the insulation.
 3. The insulator can become **polarized** by an electric field: the internal charges move very slightly to create a local dipole.

C. The capacitor

1. Definition: A **capacitor** is a set of two conductors called **armatures**, separated by an **insulator** (or dielectric).

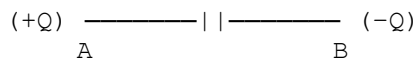
- **A:** Internal plate of the capacitor
- **B:** External plate of the capacitor

The system $[A, B]$ forms a **capacitor**.

These plates carry **charges of equal absolute value and opposite signs**:

$$Q_A = -Q_B$$

Schematic representation:



2. Capacitance of a Capacitor: When the two plates are connected by a conducting wire, a charge Q flows through the wire.

This charge is called the **charge of the capacitor**.

The **charge of the capacitor** is the one carried by the **internal plate**.

The **capacitance of a capacitor** is the quantity C such as:

$$C = \frac{Q}{V_1 - V_2}$$

where:

- Q : charge of the internal plate,
- $V_1 - V_2$: potential difference between the two plates.

👉 The **capacitance of a capacitor depends only on its geometry** (surface, distance between armatures, nature of the dielectric). **Unit:** 1 F (farad) = $1 \frac{C}{V}$

3. Energy of a capacitor: The electrostatic energy stored in a charged capacitor is given by:

$$\xi = \frac{Q^2}{2C} = \frac{1}{2} Q(V_1 - V_2)$$

This energy corresponds to the **work done** to transfer the charge Q **between the plates**.

4. Capacitor combinations: Several capacitors can be combined in different ways:

a. Series Combination: In a series connection, the internal plate of one is connected to the external plate of the next: $C_1 - C_2 - C_3 - \dots - C_n$

👉 **Characteristics:**

- All capacitors carry **the same charge Q** .
- The total potential difference is the sum of the partial differences:

$$V_{MN} = V_1 + V_2 + V_3 + \dots + V_n = Q\left(\frac{1}{C_1} + \frac{1}{C_2} + \dots + \frac{1}{C_n}\right)$$

The **equivalent capacitance** is defined as C of the parallel combination:

$$\frac{1}{C_{tot}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots + \frac{1}{C_n} = \sum_{i=1}^n \frac{1}{C_i}$$

b. Parallel combination: In a **parallel** connection, all the internal plates are connected, as well as all the external plates.

Each capacitor is subject to **the same potential difference V_{MN}** .

👉 **Characteristics:**

- $V_1 = V_2 = \dots = V_n = V_{MN}$
- The total load is the sum of the partial charges: $Q = Q_1 + Q_2 + Q_3 + \dots + Q_n$
And as $Q_i = C_i V_{MN}$, we obtain: $Q = V_{MN}(C_1 + C_2 + \dots + C_n)$

Thus, the **equivalent capacitance** of the grouping is:

$$C = C_1 + C_2 + C_3 + \dots + C_n$$

Or, in general form: $C_{tot} = \sum_{i=1}^n C_i$

c. Mixed grouping (series–parallel): It is the **combination** of the two previous types.

We begin by calculating the equivalent **capacitance of the branches in series**, then that of the **groups in parallel**, until the **total capacitance** of the circuit is obtained.

4: Electrokinetics and Electrical Circuits

I. General Introduction: Electrokinetics is the part of electricity that studies **the electrical charges in motion** within the material media (conductors, electrolytes, biological tissues).

Unlike electrostatics (immobile charges), here the **charges circulate under the action of an applied electric field**.

In the human body, this movement is ensured by **ions** (Na^+ , K^+ , Ca^{2+} , Cl^- , etc.) present in intracellular and extracellular fluids.

Fundamental principle:

The movement of free electrons creates an **electric current**, measurable and exploitable. The charges move:

- Spontaneously under the effect of an **electric field**,
- or maintained by a **generator** that creates a **potential difference (voltage)**.

II. Electric Current:

1 Definition: An **electric current** is an ordered movement of electric charges.

- In a **metal conductor**: these are the free electrons.
- In a **biological tissue**: these are the ions.

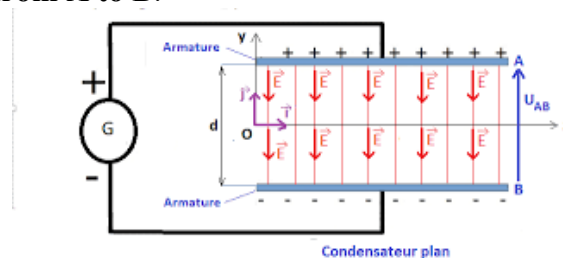
2 Nature of the electric current: In a metal conductor, the electric current results from the orderly displacement of free electrons.

 **Structure of a metal:** A **metal** is made up of **almost fixed positive ions** (crystal lattice), surrounded by **free electrons** that can easily move throughout the conductor's volume. These free electrons are responsible for **electrical conduction**.

 **Electric field in the conductor :** When applying a **potential difference** ($\Delta V = V_A - V_B$) Between the two ends of a conductor, an **internal electric field** is established:

$$\vec{E} = -\nabla V$$

 The field is **directed from high potentials into low potentials**. Thus: If $V_A > V_B$, then the field $\vec{E}$ is **oriented from A to B**.



⚡ Force on a charge

A charge q placed in this field is subjected to an electric force:

$$\vec{F} = q\vec{E}$$

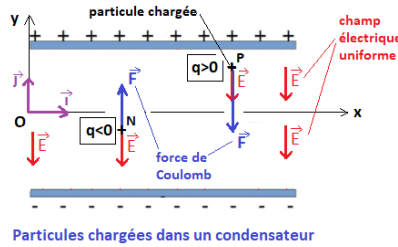
For a **positive charge**, the force is **in the same direction** as $\vec{E}$.

- For an **electron** ($q = -e$), force is **opposed** to $\vec{E}$.

Thus:

$$\vec{F}_e = -e\vec{E}$$

⇒ **Electrons move from point B (low potential) to A (high potential).**



🔄 Conventional flow direction

Element	Direction of movement
Real electrons	From B to A
Electric field	From A to B
Conventional current (I)	From A to B

⚡ So, **conventional current** has the **same direction** as the **electric field**, but **opposite** to the **actual motion** of electrons.

💡 **Energy interpretation** : The electric field provides energy to the charges:

- **Negative charges** (electrons) **gain kinetic energy** as they move up to the highest potential.
- The **positive charges** (if there were any) would spontaneously move to the lowest potential.

3 – Production of electric current:

When the armatures of a charged capacitor are connected by a conductive wire, a discharge occurs: an electric current momentarily flows through the wire.

👉 This current lasts only an instant: this is referred to as transient state.

To obtain a constant current, it is necessary to use a **generator**—a device capable of maintaining a constant potential difference (voltage) between its terminals.

⚙️ Role of the generator:

A **generator** transforms **non-electrical energy** (chemical, mechanical, solar, etc.) into **electrical energy**.

It is **the source** of current in a circuit.

Examples:

- Cell or battery → chemical energy → electrical energy.
- Alternator → mechanical energy → electrical energy.

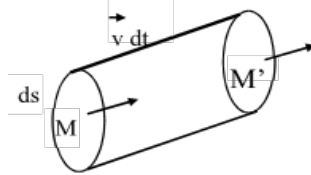
⚡ Standardized symbol: (+) ——— (-)

The pole P (+) has a higher potential than the pole N (–).
by convention, the electric current flows from the **positive pole** to the **negative pole**, while the electrons move in the **opposite direction**.

4 Electric current intensity and current density

📌 **1. The concept of current lines:** Current lines are the **trajectories** followed by **mobile electric charges** (called charge carriers) inside a conductor. These lines indicate the **local direction** of the electric current.

⚙️ **2. Microscopic definition of elemental intensity:** Let be a point **M** in a conductor, and **ds** a small surface element surrounding M, oriented according to the normal vector **$\vec{n}$** .



Si :

- ρ = volume density of mobile charges ($C \cdot m^{-3}$)
- $\vec{v}$ = average movement velocity of charges
- dt = time Interval

Then the amount of charge passing through **ds** during **dt** is: $dq = \rho (\vec{v} \cdot \vec{n}) ds dt$

Therefore, **the elementary intensity** (current through ds) is: $dI = \rho (\vec{v} \cdot \vec{n}) ds$ (1)

⚙️ 3. Current density vector:

We define the **current density vector** $\vec{j}$ by: $\vec{j} = \rho \vec{v}$

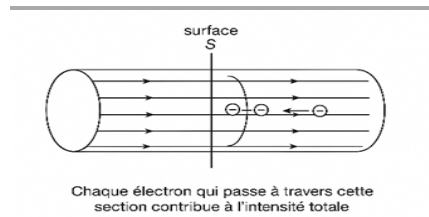
Thus, equation (1) becomes: $dI = \vec{j} \cdot \vec{n} ds$

- ♦ The vector $\vec{j}$ represents the amount of current crossing a unit of surface area.

4. General definition of current intensity

If the current passes through a surface (**S**), the total intensity **I** is the sum of the elementary intensities through each element **ds**:

$$I = \iint_S \vec{j} \cdot \vec{n} ds$$



- If S is perpendicular to the current lines: $I = \iint_S j \, ds = jS$
(The flux is uniform).

⚡ 6. Physical interpretation

- I Measures the total quantity of electricity that crosses a section of the conductor per second.
- $\vec{j}$ describes the **spatial distribution** of the current.
- If the conductor is uniform: $I = jS$
- If the current density varies in space, the surface integral is used.

■ **7. Important note:** The intensity of the electric current I also corresponds to the quantity of electricity transported per unit of time:

$$I = \frac{dq}{dt}$$

And so:

$$q = \int I \, dt$$

✚ **8. Illustrative example:** Imagine a cylindrical metal wire through which a current flows.

- ☐ The current lines are parallel to the axis of the wire.
- ☐ The surface S is perpendicular to the wire: it intercepts the current lines.
- ☐ Each electron passing through this section contributes to the total intensity.

⚡ IV – Ohm's Law

⚙ 1. Statement of Ohm's Law :

Ohm's Law relates the voltage (U) applied to a conductor at the **intensity of the current (I)** flowing through it.

At a constant temperature, the current flowing through a conductor is **proportional to the applied voltage** :

$$U = R I$$

Where:

- U : Potential Difference (volts, V)
- I : Intensity of current (amperes, A)
- R : Electrical resistance of the conductor (ohms, Ω)

🧱 2. Local (microscopic) form :

At the microscopic scale, Ohm's law is written in the form:

$$\vec{j} = \gamma \vec{E}$$

$\vec{j}$: Current density ($A \cdot m^{-2}$)

$\vec{E}$: Electric field ($V \cdot m^{-1}$)

γ : Material conductivity ($S \cdot m^{-1}$).

- ◆ The conductivity γ measures the ease with which charges move.

 **3. Resistance and resistivity:** The resistance of a conductor measures its **opposition to the passage of electric current**. It depends on the **nature of the material**, its **length**, and its **cross-section**.

 **2. General formula:** $R = \rho \frac{l}{S}$

Where:

- R : Resistance (Ω , ohms)
- ρ : Material resistivity ($\Omega \cdot m$)
- l : Conductor length (m)
- S : Conductor Section (m^2)

- ◆ **Note:** the longer the wire, $\rightarrow$ the greater the R ; the wider the cross-section, $\rightarrow$ smaller R .

3. Factors Influencing Resistance

Factor	Influence on R	Explanation
Temperature	$\uparrow T \Rightarrow \uparrow R$ (for metals)	Collisions increase between electrons and atoms
Material	depends on the resistivity ρ	Copper: good conductor; Glass: Insulating
Dimensions	$R \propto l / S$	The longer and thinner the thread, the more resistant it will be

 **4. Conductance :** Conductance is the measure of how easily an electric current flows through a conductor.

It's the opposite of resistance. (Unit : siemens (S).)

$$\sigma = \frac{1}{R}$$

 **Numerical example:** An aluminum wire of a length $l = 3 \text{ m}$ and a section $S = 2 \text{ mm}^2$ has a resistivity: $\rho = 2.8 \times 10^{-8} \Omega \cdot m$.

$$R = \rho \frac{l}{S} = 0.042 \Omega$$

- ◆ The resistance is **0,042 ohm**, very low — therefore a good conductor.

 Biological tissues are **poor conductors**, because their conductivity depends on the **presence of ions** (Na^+ , K^+ , Cl^- , etc.).

 **4. Energy interpretation:** The electrical energy transferred by the current for a time t is:

$$W = UIt = RI^2t$$

Some of this energy is **converted into heat** $\rightarrow$ Joule effect.

🦋 5. Biological Application: Membranes and Electrolytes

Biological **membranes** can be likened to **electrical resistors** (and capacitors).

- **Ionic current** across a membrane follows a law analogous to **Ohm's law**:

$$I = g(V_m - E_{ion})$$

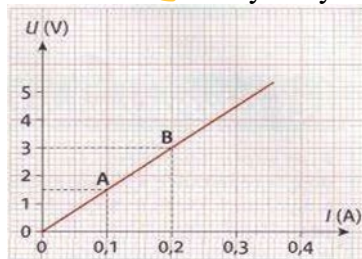
Where g is the **conductance** (S) and E_{ion} the **potential for equilibrium**.

- **Excitable cells** (neurons, muscle fibers) have **ion channels** whose conductance depends on the voltage → modified Ohm model.

✖ 6. Graphique caractéristique :

◆ **Definition:** The **characteristic of a dipole** represents the relationship between the **voltage U** applied to its terminals and the intensity I of the current that flows through it.
→ It is traced in the **plan (U, I)**.

- ◆ **Case 1: Ohmic Materials** 🖐 They obey Ohm's law: $U = R \times I$



The graph is a **straight line passing through the origin**, with a slope R .

Examples:

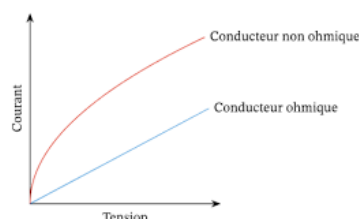
- Metals (Cu, Al).
- Electrical resistors.
- Ohmic conductors.

■ Graphic interpretation:

- The slope of the line = R
- If the line is steeper → greater resistance.

- ◆ **Case 2: Non-ohmic materials**

🖐 They **don't respect Ohm's law**: the relationship $U = RI$ is not linear.
The graph is **not a straight line**.



Examples:

- Diodes (current only passes in one direction)
- Incandescent lamps (resistance varies with temperature)
- Biological membranes (variable ion channels)

7. Practical applications

- **Linear regime:** $U = RI \rightarrow$ Constant resistance.
- **Non-linear regime:** R varies with U or I .
- **Biology:** The resistance of human skin varies greatly depending on humidity.

 **Application exercise (bioelectric):** A cell membrane can be modeled by a resistor $R_m = 50 \text{ M}\Omega$ (or $50 \times 10^6 \Omega$).

We apply a potential difference $U_m = 70 \text{ mV}$.

1. Calculate the ionic current across the membrane.
2. Express this current in nanoamperes (nA).

✓ Solution :

1. $I = \frac{U_m}{R_m} = \frac{70 \times 10^{-3}}{50 \times 10^6} = 1.4 \times 10^{-9} \text{ A}$.
2. $I = 1.4 \text{ nA}$.

 **Result:** The ionic current through a biological membrane is extremely low (in the nanoampere range).

VI – Joule's Law

♦ **1 Definition:** When an electric current passes through a conductor, part of the electrical energy is transformed into heat.

This phenomenon is called the Joule effect.

 It is a conversion of electrical energy into thermal energy due to the resistance of the conductor.

Examples:

- The filament of an incandescent lamp heats up.
- The resistors of water heaters or irons that release heat.

♦ **2 Mathematical expression:**

♦ a) Instant form: $P = U \times I$

The electrical power dissipated in the form of heat (in watts, W).

♦ b) With Ohm's Law: $P = RI^2 = \frac{U^2}{R}$

Where:

- P : Power dissipated (W, watts)
- U : Tension (V)
- I : Intensity (A)
- R : Resistance (Ω)

→ The greater the resistance or intensity, the greater the heating.

♦ **3 Energy dissipated for a time t**

$$W = P \times t = RI^2t$$

- W : energy dissipated (in joules, J)
- t : duration of the current passage (in seconds, s)

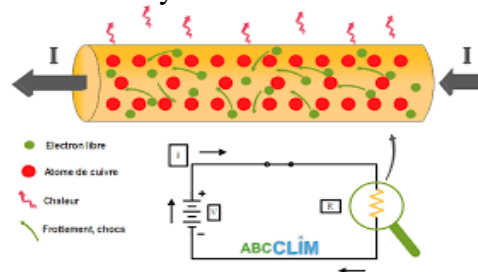
♦ **4 Numerical example:** A resistance conductor $R = 10 \Omega$ is crossed by a current $I = 2 \text{ A}$ during $t = 5 \text{ s}$.

$$W = RI^2t = 10 \times (2)^2 \times 5 = 200 \text{ J}$$

→ **200 joules** of energy are dissipated as heat.

♦ **5 Energy interpretation**

- The electric current encounters an **opposition (resistance)** to the displacement of charges.
- The kinetic energy of the electrons is **converted into thermal agitation** of the ions in the metal lattice.
→ This agitation is manifested by an **increase in the temperature** of the conductor.



♦ **6 Biological Application:** In living tissues:

- The electric current also causes **heating** proportional to the conductivity and intensity of the current.
- This effect is **used** (e.g. in diathermy, to locally heat a tissue).
- But it is also **dangerous** in the case of accidental passage of the current → internal burns or cardiac fibrillation.

 **Exercise – Biological Application:** During muscle stimulation, a resistance electrode $R = 1000 \Omega$ is subjected to a voltage $U = 5 \text{ V}$ during $t = 1 \text{ s}$. Calculate:

1. The current that runs through it;
2. The energy released by the Joule effect.

✓ **Solution :**

1 $I = \frac{U}{R} = \frac{5}{1000} = 0.005 \text{ A} = 5 \text{ mA}$

2 $W = RI^2t = 1000 \times (0.005)^2 \times 1 = 0.025 \text{ J}$

♦ **The result:** low energy but enough to **stimulate a muscle fiber** without damaging it.

⚙ **5. Technical Application: Fuse**

A fuse protects an electrical circuit. It is designed to melt when the heat due to the Joule effect exceeds a certain limit. Thus, it interrupts the current and prevents overheating.

II. Parallel connection

♦ **1. Definition:** Resistors are in **parallel** when subjected to the **same voltage**. The current is divided between the branches.

♦ 2. Law of currents

$$I = I_1 + I_2 + I_3$$

And as $I = \frac{U}{R}$:

$$\frac{U}{R_{eq}} = \frac{U}{R_1} + \frac{U}{R_2} + \frac{U}{R_3} \Rightarrow \frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

Note: The equivalent resistance in parallel is **smaller** than the smallest of the resistors.

💡 **Numerical example:**

Two resistances $R_1 = 6 \Omega$ and $R_2 = 3 \Omega$: $\frac{1}{R_{eq}} = \frac{1}{6} + \frac{1}{3} = \frac{3}{6} \Rightarrow R_{eq} = 2 \Omega$

♦ The result: the equivalent resistance is lower than 10Ω .

III. Mixed connection: A **mixed** combination combines series and parallel assemblies. The circuit is simplified step by step until a single equivalent resistor is obtained.

💡 **Example:** Two resistances $R_2 = 4 \Omega$ and $R_3 = 12 \Omega$ are in parallel, all in series with $R_1 = 6 \Omega$.

$$\frac{1}{R_{23}} = \frac{1}{4} + \frac{1}{12} = \frac{4}{12} \Rightarrow R_{23} = 3 \Omega$$

$$R_{eq} = R_1 + R_{23} = 9 \Omega$$

🔌 VIII. Generator:

The **generator** is a device that raises the potential of electrical loads: it receives charges from one terminal and returns them from the other terminal with a higher potential. In the **external circuit**, the **conventional current** flows from the positive (+) terminal to the negative terminal (-).

Inside the generator, non-electrostatic forces (e.g. chemical) move the charges from the **negative** pole towards the **positive** pole, i.e. from the lowest potential to the highest potential.

A real generator has an **internal resistance** r that opposes the movement of charges and causes losses due to the Joule effect.



⚡ VIII. Electromotive force (e.m.f.)

♦ **VIII.1 General definition** La force électromotrice (e.m.f.), also called electromotance, is a characteristic parameter of an electric generator.

Despite its name, it is not a force in the mechanical sense, but a quantity of the same nature as voltage.

It represents the energy provided by the generator per unit load to move a positive charge through the generator, from the negative (-) pole to the positive (+) pole.

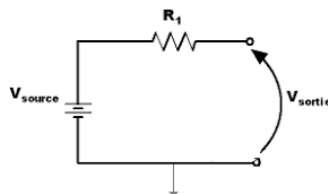
$$\mathcal{E} = \frac{W}{q}$$

with:

- $\mathcal{E}$: e.m.f. in **volts (V)**
- W : work provided by the generator (in joules, J)
- q : displaced charge (in coulombs, C)

👉 **Unit:** 1 volt = 1 joule / coulomb

♦ **VIII.1.1 Open Circuit:** When the circuit is **open**, there is no **current** ($I = 0$). The **potential difference** between the terminals of the generator is then **equal to its e.f.m.**, because the internal resistance does not cause any voltage drop.

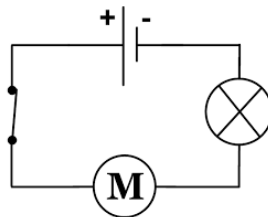


$$\mathcal{E} = U = V_A - V_B \text{ (en volts)}$$

Note: The internal resistance r of the generator does not play a role in this case.

→ The voltage measured between the terminals corresponds to the **"open-circuit" voltage** (or "no-load" voltage).

♦ **VIII.1.2 Closed Circuit:** When the generator powers a circuit (**closed circuit**), a current of intensity **I** flows. Part of the e.m.f. is then "lost" in the **internal resistance r** of the generator.



$$\mathcal{E} = (V_A - V_B) + rI$$

- $V_A - V_B = U$: Useful voltage at generator terminals.
- rI : internal voltage drops (due to the Joule effect in the internal resistance)

♦ **VIII.1.3. Electrical power relationship**

⚡ Total power supplied by the generator: $P = \mathcal{E}I$

⚡ Power dissipated by the Joule effect in the internal resistance: $P_{dissipated} = rI^2$

⚡ Usable power (available for outdoor circuit): $P' = P - P_{dissipated} = (V_A - V_B)I$

💡 **Energy interpretation**

The generator transforms energy (chemical, mechanical, solar, etc.) into **electrical energy**

- Part of it is transmitted to the external circuit (**useful**).
- Some of it is dissipated as **internal heat** (Joule effect).

⚙️ **IX. Receivers:** A receiver converts the electrical energy received into another form of energy (mechanical, chemical, thermal, etc.).

It is associated with a **back-electromotive force (b.e.m.f.)**, which represents the voltage opposing the flow of the current.

IX.1 Open Circuit: If the receiver is in open circuit, the ddp at its terminals is:

$$e = U = V_A - V_B \text{ (volts).}$$

IX.2 Closed circuit: For a receiver with back-electromotive force (b.e.m.f.) and internal resistance r , crossed by a current I , the terminal voltage satisfies:

$$V_A - V_B = e + r I.$$

Electrical energy supplied $(V_A - V_B)I$ is used for:

- Providing useful energy eI (mechanical, chemical, ...),
- Producing Joule Losses rI^2 .

Thus:

$$(V_A - V_B)I = eI + rI^2.$$

X. Kirchhoff's laws: Kirchhoff's Laws allow for the analysis of networks (complex circuits) containing multiple generators and receivers.

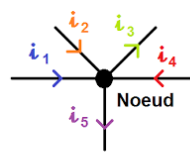
X.1 Definitions:

- **Network:** a set of conductive connections linking generators and receivers via resistors.
- **Node:** a point where at least three circuit elements meet.
- **Branch:** a set of elements in series between two nodes.
- **Loop:** a succession of branches forming a closed circuit.

X.2 Node law (1st law)

The algebraic sum of the currents arriving at a node is equal to the sum of the currents leaving it:

$$\sum I_{\text{incoming}} = \sum I_{\text{outgoing}}.$$



Loi des noeuds :

$$i_1 + i_2 + i_4 = i_3 + i_5$$



Physical interpretation: A **node** is a connection point between several conductors. Since electric charges **cannot accumulate in a node** (principle of conservation of charge), the amount of current **entering** the node must be **equal** to the amount of current **leaving it**.

X.3 Loop Law (2nd Law)

For a loop, the algebraic sum of the voltage drops is equal to the algebraic sum of the e.m.f. encountered:

$$\sum R_i I_i = \sum \mathcal{E}_j \mathcal{E}_j$$

where the sign $\varepsilon_j = \pm 1$ depends on the chosen direction of travel:

- $\varepsilon_j = +1$ if, while traversing, exits through the terminal (+) of the generator (we find the e.m.f. in the direction of "engine"),
- $\varepsilon_j = -1$ otherwise.

Practical rule: Choose a direction of travel and apply Ohm's law and the addition of voltages while taking signs into account.

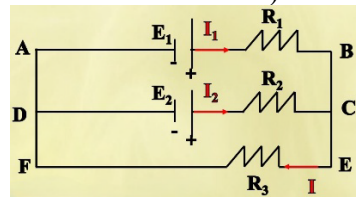
XI. Applications

1) Circuit resolution (example)

For the given circuit (two generators E_1, E_2 and resistors R_1, R_2, R_3), we write the equations of the lattices and the knot. With the following values:

Input values $E_1 = 5 \text{ V}, E_2 = 4 \text{ V}, R_1 = R_3 = 1 \Omega, R_2 = 2 \Omega,$

The equations are (depending on the direction chosen):



$$\text{Loop } ABCDA: R_1 I_1 - R_2 I_2 = E_1 - E_2,$$

$$\text{Loop } ABEFA: R_1 I_1 + R_3 I = E_1,$$

$$\text{Node } C: I_1 + I_2 = I.$$

We denote: I_1 Left-hand loop current, I_2 right-hand loop current, and I current in the common branch R_3 .

1) Loop equations

Loop $ABCD$ (the loop containing E_1, R_1, R_2, E_2 according to the statement):

$$R_1 I_1 - R_2 I_2 = E_1 - E_2.$$

Loop $ABEFA$ (the loop that contains E_1, R_1, R_3): $R_1 I_1 + R_3 I = E_1$.

2) **Node equation C (Node law) at the node C:** $I_1 + I_2 = I$.

3) **Numerical substitution and resolution** let's substitute the values into the equations:

$$\text{Loop } ABCDA: I_1 - 2I_2 = 1$$

$$\text{Loop } ABEFA: I_1 + I = 5$$

Node C: $I = I_1 + I_2$. Let's replace I in the second equation: $\Rightarrow 2I_1 + I_2 = 5$.

We therefore obtain the simple system :

$$\begin{cases} I_1 - 2I_2 = 1, \\ 2I_1 + I_2 = 5. \end{cases}$$

Results (final)

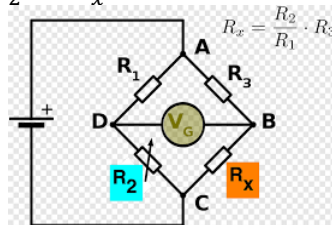
$$I_2 = 0,6 \text{ A} \quad I_1 = 2,2 \text{ A} \quad I = 2,8 \text{ A}$$

2) Wheatstone Bridge: The Wheatstone Bridge is an experimental device used to determine the value of an unknown resistance R_x by comparison.

It is made up of four resistors R_1, R_2, R_3, R_x forming a quadrilateral in which two opposite vertices are connected to a voltage source, and the other two to a galvanometer.

At the equilibrium of the bridge, the potential of the two points connected to the galvanometer is identical, the current flowing through it is zero, and the equilibrium condition is written :

$$\frac{R_1}{R_2} = \frac{R_3}{R_x} \text{ So } R_x = R_3 \times \frac{R_2}{R_1}$$



Bioelectricity — Membrane Potential and Action Potential

Introduction

Bioelectricity studies the electrical phenomena produced by living cells. It is based on the asymmetric distribution of ions (Na^+ , K^+ , Cl^-) either side of the membrane plasmic, at the origin of the **resting potential**. When stimulation occurs, this distribution changes, generating an **action potential** that propagates along the excitable membranes, ensuring the **transmission of nerve impulses**.

The cell, the structural and functional basic unit of any living organism, is delimited by a **cell membrane** that separates the intracellular environment from the extracellular environment.

➔ This electrical property of the membrane is essential for proper cellular function. À l'état stationnaire, la cellule présente un **potentiel de repos négatif**, la face interne étant plus négative que la face externe.

I. Cell membrane

I-1. Definition of the cell membrane

The **plasma membrane**, also known as **the cell membrane**, is a thin biological structure that separates the intracellular environment (cytoplasm) from the extracellular environment.

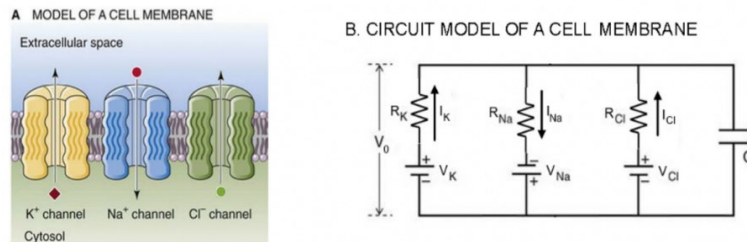
It ensures the **protection**, **maintenance of integrity** and regulation of **exchanges** between the cell and its environment.

Although its plasticity **is sometimes referred** to as its flexibility, and its ability to deform, the correct term remains **"plasma membrane"** (and not "plastic membrane"). This property gives the cell functional **flexibility** essential for communication, ion transport and maintenance of membrane potential.

I-2. Electrical Model of the Cell Membrane

The membrane can be represented by an **equivalent electrical model**. In this model, we distinguish:

- C_m , the capacity of the membrane, approximately $1 \mu\text{F} \cdot \text{cm}^{-2}$, proportional to the surface area of the membrane;
- Ionic conductances (symbolized by resistances), generally variable over time depending on the opening or closing of ion channels;
- Electrochemical batteries, corresponding to the equilibrium potentials of the different ions (given by Nernst's equation);
- A voltage generator, representing active membrane pumps, including the sodium-potassium pump (Na^+/K^+ -ATPase), that maintains ion gradients.



II-3. Membrane potential

Definition: The membrane potential corresponds to the difference in electrical potential between the intracellular and extracellular media.

It results from an uneven distribution of ions (Na^+ , K^+ , Cl^- , etc.) on either side of the cell membrane.

It is **negative** when the inner side of the membrane is more charged with negative ions than the outer side, and vice versa.

II-3-1. Resting Potential : The resting potential is the electrical potential difference between the inside and the outside of the cell in the absence of any nerve impulse.

Typical values:

- Neuron: **-70 mV**
- Skeletal muscle fiber: **-90 mV**

Origin of the resting potential

1. **Asymmetric ion distribution (Donnan phenomenon):** The unequal distribution of diffusible ions (Na^+ , K^+ , Cl^-) is due to the presence of non-diffusible ions (proteins, organic anions).

The Donnan phenomenon describes this passive equilibrium, generating a transmembrane potential difference while maintaining the electroneutrality of both compartments.

2. **Selective permeability of the membrane:** The plasma membrane acts as a selective barrier, allowing only certain ions to pass through.

This selective permeability allows the cell to maintain an internal ionic composition different from that of the extracellular medium.

Physiological role: The resting potential is **essential** for the functioning of **excitable cells** (neurons, muscle fibers, heart cells).

In these cells, it conditions the **genesis of the action potential**.

Example:

In myocardial cells, the ions K^+ predominate inside, and the ions Na^+ on the outside, which creates the resting potential difference.

II-3-2. Potentiel d'action (PA)

Definition: The action potential is a rapid and transient variation of the membrane potential of an excitable cell (neuron or muscle cell), characterized by depolarization followed by repolarization. It is triggered when a stimulation reaches a specific threshold, causing the opening of voltage-gated ion channels. This disturbance allows for the transmission of electrical signals along the membrane.

Fundamental property: The action potential obeys the all-or-none law: if the stimulus reaches the threshold, the AP is triggered with maximum amplitude; below the threshold, no AP is generated.

Action potential phases

1. Rest phase: Stable membrane potential (approximately -70 mV in a neuron). The cell is polarized.

2. Depolarization: The membrane suddenly becomes permeable to Na^+ ions → Massive entry of Na^+ into the cell → The membrane potential becomes positive (polarity reversal).

3. Repolarization: Closing of Na^+ channels and opening of K^+ channels → Exit of K^+ from the cell → Return to negative values of the potential

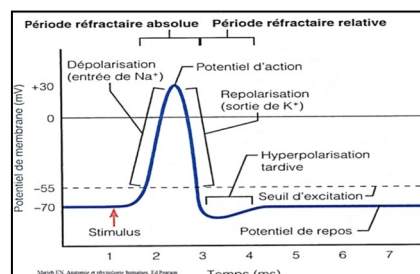
4. Post-potentiel d'hyperpolarisation: The membrane temporarily becomes more negative than the resting potential → Due to the persistent high permeability to K^+ .

5. Return to the resting potential: Restoration of ion concentrations by the pump Na^+/K^+ ATP-dependent → The cell becomes excitable again.

Trigger mechanism: When an excitable cell receives sufficient stimulus:

- The resting potential is disrupted;
- The membrane suddenly becomes permeable to Na^+ ;
- A depolarization followed by a repolarization occurs successively.

Functional importance : The action potential constitutes the elementary basis of electrical communication in the nervous and muscular systems, allowing for the rapid propagation of information along cellular membranes.



♦ II-3. Ion Channels

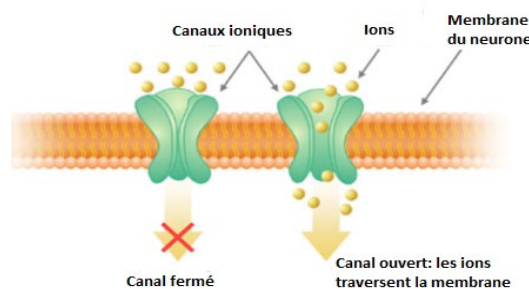
Definition: Ion **channels** are protein pores embedded in the plasma membrane.

They allow for the **rapid and selective passage** of ions between the **intracellular** and **extracellular** environments.

They are made of **protein subunits** that assemble to form a **cylindrical channel**, allowing for the circulation of water and small ions.

Depending on their regulation, these channels can be:

- **Open** (allowing ions to pass through),
- **Closed** (no passage),
- Or **constantly open**, called **leakage channels (pores)**.



■ Schematically, an ion channel is comparable to a door between two environments.

II-3-1. Functions and electrical role

Ion channels ensure the selective passage of ions, which is responsible for the generation of membrane electrical potentials.

The membrane, normally impermeable to ions, becomes permeable thanks to these channels.

Thus:

- The opening or closing of channels determines ionic currents,
- These ionic movements create potential differences between the two sides of the membrane.

→ **En résumé :** les canaux ioniques contrôlent la perméabilité membranaire et sont à l'origine des phénomènes électriques cellulaires.

♦ II-4. Migration phenomena

Definition: Electrical migration corresponds to the displacement of ions under the influence of an electrical field (E).

This displacement occurs with a decrease in the ions' potential energy.

Characteristics

- Existence of an electrical force applied to the ions (Coulomb forces).
- Requires an energy proportional to this force and the displacement.
- Concerns only charged species (anions or cations).
-

Expression of the electrical molar flow rate:

$$J_e = -zFbSC_m \text{ grad}V$$

With:

- J_e : Electrical molar flow rate (mol/s)
- z : Valence of the ion
- F : Faraday's constant (96500 C/mol)
- b : Molar mechanical mobility (s/kg)
- S : Diffusion surface area (m²)
- C_m : Molar concentration
- $\text{grad}V$: Electrical potential gradient

We can also write: $J_e = -\mu S C_m \text{grad} V$

With: $\mu = z F b e t \mu = \frac{z D F}{R T}$

where:

- D : diffusion coefficient (m^2/s)
- R : ideal Gas Constant
- T : absolute temperature (K)

II-5 Electrical conductivity

Electrical conductivity (γ): Represents the ease with which charge carriers (electrons or ions) move electrical current through a conductive medium.

If a current I circulates from **A** to **B**, separated by a distance l , we have:

$$V_A - V_B = R I = E l$$

Where E is the electric field and R conductor resistance.

The **density of the current** is defined by: $j = \frac{I}{S}$

with S : the cross-section area of the conductor.

Where: $R j S = E l \Rightarrow j = \frac{E l}{R S} = \gamma E$

We thus obtain:

$$\gamma = \frac{l}{R S}$$

The quantity γ is called **electrical conductivity**, and its inverse $\rho = 1/\gamma$ is the **electrical resistivity** (en $\Omega \cdot \text{m}$).

II-6 Electric mobility : Electric mobility (μ) is defined as the velocity acquired by a charged particle under the action of a unit electrical field:

$$v = \mu E \Rightarrow \mu = \frac{v}{E}$$

It is expressed in:

$$[\mu] = \text{m}^2 \text{V}^{-1} \text{s}^{-1}$$

The **electrical conductivity** of a medium is linked to the charge and mobility of the carriers by the relationship:

$$\gamma = n q \mu$$

Where:

- n : number of charge carriers per unit volume,
- q : electrical charge of a carrier,
- μ : mobility of the carriers.

In an electrolytic solution, the carriers are ions:

$$\gamma = \sum n_+ q_+ \mu_+ + \sum n_- q_- \mu_-$$

With:

- n_+, n_- : Number of cations and anions,
- μ_+, μ_- : Ionic mobilities of cations and anions.

II-7 Electrical conductivity of an ionic solution: The electrical conductivity of an ionic solution is given by : $\gamma = F C_m (\mu_+ + \mu_-)$

Where:

- γ : Electrical conductivity,
- F : Faraday's constant ($96\,500 \text{ C} \cdot \text{mol}^{-1}$),
- C : Molar concentration of the solution,
- μ_+, μ_- : ionic mobilities of positive and negative ions.

Demonstration :

According to **Ohm's Law** : $\boxed{U = RI}$

The resistance of a conductor (Here is the solution) is: $R = \frac{\rho l}{S}$
with ρ resistivity, l the thickness and S the surface of the electrodes.

Thus: $U = \frac{\rho l}{S} I \Rightarrow \rho = \frac{US}{lI}$

Conductivity is the inverse of resistivity: $\boxed{\gamma = \frac{1}{\rho} = \frac{lI}{US} = \frac{J}{E}} \quad (1)$

where J is the density of the current and E is the electric field.

However, **mobility** is: $\boxed{v = \mu E}$

The current density corresponds to the total charge transported: $J = nezv = nez\mu E$

The solution contains positive and negative ions. For a molar solution: $FC = nez$

Thus: $\boxed{J = FC(\mu_+ + \mu_-)E} \quad (2)$

By comparing (1) and (2), we obtain: $\boxed{\gamma = FC(\mu_+ + \mu_-)}$

Application – Example:

Let us consider a solution of hydrochloric acid (HCl) of concentration $C_m = 10 \text{ mmol/L}$.

The ionic mobilities of the ions are: $\mu(\text{H}^+) = 35 \text{ } \mu\text{m/s}$ and $\mu(\text{Cl}^-) = 7,63 \text{ } \mu\text{m/s}$

When they are subjected to an electric field of $E = 1 \text{ V/cm}$ at the temperature of 25°C .

Knowing that Faraday's constant is $F = 96500 \text{ C/mol}$,

Calculate:

- Ionic mobilities expressed in $\boxed{\text{m}^2 \cdot \text{s}^{-1} \cdot \text{V}^{-1}}$.
- Conductivity γ of the solution.
- Resistivity ρ of the solution.

✓ **Solution** : Data

$$E = 1 \text{ V/cm} = 100 \text{ V/m}$$

$$v_{\text{H}^+} = 35 \text{ } \mu\text{m/s}$$

$$v_{\text{Cl}^-} = 7.63 \text{ } \mu\text{m/s}$$

$$C_m = 10 \text{ mmol/l} = 10 \text{ mol/m}^3$$

$$F = 96500 \text{ C/mol}$$

1) Mobility $\mu = v/E$ (en $\text{m}^2 / (\text{V} \cdot \text{s})$) $\mu_{\text{H}^+} + \mu_{\text{Cl}^-} = 4,263 \times 10^{-7} \text{ m}^2 / (\text{V} \cdot \text{s})$

2) Conductivity : $\gamma = F C (\mu_+ + \mu_-) = 0,411 \text{ S} \cdot \text{m}^{-1}$

3) The associated resistivity is $\rho = 1/\gamma \approx 2,43 \text{ } \Omega \cdot \text{m}$

II-8 — DONNAN EQUILIBRIUM:

Principle (reminder):

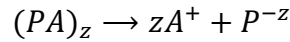
When a semi-permeable membrane separates two compartments I and II, and that a charged macromolecule (for example, a protein) is present in only one of the two, the small diffusible ions distribute themselves to ensure both local electroneutrality and the equality of electrochemical potentials at equilibrium.

This distribution leads to different ion concentrations on either side of the membrane and a nonzero membrane potential difference: this is the **Donnan effect**.

Problem description (modeled) :

"Two compartments, I and II, separated by a membrane permeable to small ions A^+ and B^- , but impermeable to the protein, initially contain these ions at the same concentration C .

A protein is introduced into compartment I $(PA)_z$ of concentration C' which is dissociated according to:



The protein P^{-z} is immobilized (it does not pass through the membrane).

In equilibrium, we note C_m the concentration (in $\text{mol} \cdot \text{m}^{-3}$ or $\text{mol} \cdot \text{L}^{-1}$, (depending on the chosen units) of the small ions that have diffused from the compartment I to the compartment II.

The aim is to determine the ionic concentrations at equilibrium in I and II, as well as the expression of membrane potential $V_2 - V_1$ corresponding to **the Donnan effect**.

a) Equilibrium concentrations (As a function of C, C', z, C_m)

After adding and distributing $C_m(\text{net})$ of ions from I to II:

- **Compartment I:** $[A^+]_I = C + zC' - C_m, [B^-]_I = C - C_m.$
- **Compartment II:** $[A^+]_{II} = C + C_m, [B^-]_{II} = C + C_m.$

Verification of electroneutrality:

In I: $[A^+]_I = [B^-]_I + zC'$

In II: $[A^+]_{II} = [B^-]_{II}$

Table of concentrations

Compartment	Initial state	End state
I	$[A+]_I = C + zC'$	$[A+]_I = C + zC' - m$
	$[B-]_I = C$	$[B-]_I = C - C_m$
	$[P^{-z}]_I = C'$	$[P^{-z}]_I = C'$
II	$[A+]_{II} = C$	$[A+]_{II} = C + C_m$
	$[B-]_{II} = C$	$[B-]_{II} = C + C_m$

b) Equality of electrochemical potentials — Nernst's Law

For an ion i of charge z_i , the equality of electrochemical potentials between I and II At equilibrium, it is written as:

$$RT \ln \frac{[i]_I}{[i]_{II}} + z_i F (V_1 - V_2) = 0$$

- For A^+ ($z_A = +1$):

$$V_2 - V_1 = \frac{RT}{F} \ln \frac{[A^+]_I}{[A^+]_{II}} = \frac{RT}{F} \ln \frac{C + zC' - C_m}{C + C_m}$$

- For B^- ($z_B = -1$):

$$V_2 - V_1 = -\frac{RT}{F} \ln \frac{[B^-]_I}{[B^-]_{II}} = -\frac{RT}{F} \ln \frac{C - C_m}{C + C_m}$$

By equating the two expressions, the potential difference can be eliminated to determine C_m .

c) Determination of C_m

By setting:

$$\frac{C + zC' - C_m}{C + C_m} = \frac{C + C_m}{C - C_m}$$

and by simplifying it, we get:

$$C_m = \frac{C C' z}{4C + C' z}$$

C_m Represents the net concentration of ions that have migrated from I to II at equilibrium.

d) Useful Final Expressions

- **Equilibrium concentrations:**

$$[A^+]_I = C + zC' - \frac{CC'z}{4C + C'z}; [B^-]_I = C - \frac{CC'z}{4C + C'z}$$

$$[A^+]_{II} = [B^-]_{II} = C + \frac{CC'z}{4C + C'z}$$

- **Donnan potential difference:**

$$V_1 - V_2 = \frac{RT}{F} \ln \left(\frac{C + zC' - C_m}{C + C_m} \right)$$

$$\left(\text{À } 25^\circ\text{C}, \frac{RT}{F} \approx 0.0257 \text{ V} \right)$$

e) Limiting case (intuition)

- If $C' = 0 \rightarrow C_m = 0$ et $V_2 - V_1 = 0$: no Donnan effect.
- If $C'z \ll C \rightarrow$ approximation:

$$C_m \approx \frac{CC'z}{4C} \approx \frac{zC'}{4}$$

For low protein concentrations, C_m is proportional to C' and independent of C To the first order.

- If $C'z \gg C$, C_m tends towards C , indicating a significant redistribution (validate the model's consistency in this extreme case).

Application exercise (Calculation of the Nernst potential):

We consider a potassium ion K^+ through the membrane of a cell.

The ionic concentrations are:

$$[K^+]_{\text{int}} = 140 \text{ mM and } [K^+]_{\text{ext}} = 5 \text{ mM.}$$

Body temperature is $T = 310 \text{ K}$.

Question :

Calculate the equilibrium potential (Nernst potential) of the ion K^+ .

For the K^+ :

$$[K^+]_{int} = 140 \text{ mM}, [K^+]_{ext} = 5 \text{ mM}, T = 310 \text{ K}.$$

$$E_K = \frac{RT}{zF} \ln \frac{[K^+]_{ext}}{[K^+]_{int}} = \frac{(8.314 \times 310)}{96485} \ln \frac{5}{140} = -89 \text{ mV}$$

✅ **Results:** $E_K = -89 \text{ mV}$.

Important notes

- Each solution remains electrically neutral at equilibrium.
- The sign of potential depends on the sign of the protein:
the compartment containing the protein has the same sign as the protein itself.
- The Donnan effect is only observable through a **dialysis membrane**.
- Ions must move by **passive transport**.

For each ion, the Donnan relation is written as:

$$V_2 - V_1 = -\frac{RT}{zF} \ln \frac{[A^+]_{II}}{[A^+]_I} = -\frac{RT}{zF} \ln \frac{[B^-]_I}{[B^-]_{II}}$$

The Donnan ratio is defined as:

$$r = \frac{[A^+]_{II}}{[A^+]_I} = \frac{[B^-]_I}{[B^-]_{II}}$$

MEDICAL APPLICATIONS OF ELECTRICITY

I. Diagnostic applications:

At the cellular level: resting and action membrane potentials:

Every living cell exhibits an **electrical potential difference** between the inner and outer faces of its plasma membrane.

La traduction de ce passage est:

In most cells, this potential difference remains substantially constant. However, muscle cells and especially **neurons** can modify their electrical activity, either:

- ✚ Locally (**evoked potential**, local excitability),
- ✚ Either in a propagated manner (**nerve impulse**), thanks to an orderly **control of their membrane permeability** to Na^+ et au K^+ .

The cell membrane behaves as a dielectric with a relative permittivity close to that of lipids, forming an insulating layer between two conductive media. For the **electrical characterization** of biological media, it is necessary to use **measurement devices** and models to extract the conductivity and **permittivity** parameters.

Different measurement methods can be applied depending on the frequency studied. Thus, there exist several models of varying complexity to represent the measurement setup and the sample under test.

It has been possible to measure the intra- and extracellular electrical activity of cells cultured *in vitro* using electrodes with dimensions comparable to those of the cells.

This advance has paved the way for numerous studies seeking to measure extracellular activity using electrical techniques derived from micro-technologies.

An electrode is a device used to measure the electrical potential difference between two points in a biological system.

I.1 TECHNIQUES FOR MEASURING POTENTIALS

I-1.1 La notion d'électrode impolarisable

If **metal electrodes** were simply placed in contact with a **biological medium**, the resulting measurements would be **distorted**.

Indeed, biological media are **electrolytes** that react chemically with the electrodes, creating **electrochemical cells**.

The inherent **potentials** of these cells interfere with the potentials to be measured: it is then said that the **electrodes become polarized** upon contact with an electrolytic medium.

→ To avoid this polarization, we use **unpolarizable electrodes**, made up of a **succession of conductive media** chosen so that their own potential does **not disturb** the measurement. The following combination is most often used:

Metallic silver - Silver chloride - Sodium or potassium chloride
Ag – AgCl – NaCl or KCl

I-1-2. Microelectrodes and surface electrodes

There are two main categories of measurements of **electrophysiological potentials**:

1. Measurements at the cellular or tissue level:
→ Very fine electrodes or microelectrodes capable of penetrating inside cells are used to measure the membrane potential.
2. Macroscopic measurements:
→ Made with surface electrodes placed on the skin. Electrical contact between the skin and the electrode is provided by a conductive gel (often saturated with NaCl), which constitutes the third element of the unpolarizable sequence.

Thus, in both cases, an **unpolarizable electrode is obtained**.

I-2. Recording of electrical activity at a single point: **Electrocochleography (ECoG)**

Electrocochleography is a technique allowing the recording of cochlear electrical potentials. It is used to study hearing abnormalities by recording the currents generated by the cochlear nerve cells of the inner ear.

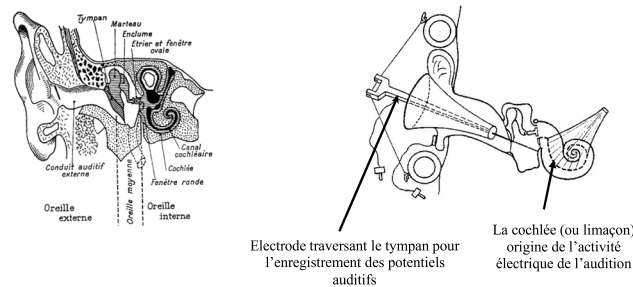
🧠 Principle:

L'ECoG enregistre les potentiels générés par:

- The organ of Corti,
- The nerve fibers of the auditory nerve, in response to sound stimulation.

✂ Method:

A microelectrode is placed transtympanic, under local anesthesia, on the promontory (near the round window) according to the method described by *Aran and al.*



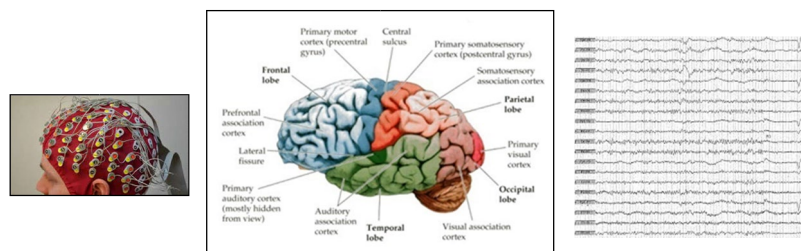
🧠 I-3. Recording of cerebral electrical activity at a large number of measurement points:

Electroencephalogram (EEG)

An electroencephalogram (EEG) is an examination that measures and records the electrical activity of the brain.

Electrodes are attached to the scalp and connected to a computer via cables.

The signals obtained reflect the electrical currents of cortical neurons, corresponding to different brain states (wakefulness, sleep, epilepsy...).



❤ I-4. Recording of cardiac electrical activity

Electrocardiogram (ECG): It is a non-invasive medical examination that records the electrical activity of the heart using electrodes placed on the skin, allowing for the assessment of heart rhythm and electrical conduction, and the detection of potential abnormalities such as:

- Arrhythmia: a heart rhythm disorder where the heart beats too fast, too slow, or irregularly.
- Ischemia is a decrease in blood supply to an organ—in this case, the heart—which can lead to a lack of oxygen in the tissues.

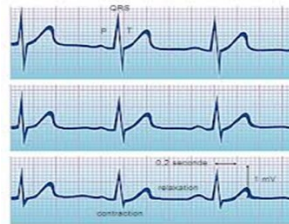
The electrocardiogram studies:

- Atrial and ventricular activity, using electrodes placed on the chest, wrists, and ankles.

The examination is quick, painless, and can be performed:

- In consultation,
- In a hospital setting,
- At home,
- or in an emergency (in a public setting).

The electrodes are applied to skin coated with conductive gel, and the signals are recorded on graph paper.



Interpretation of the main waves of the electrocardiogram:

The electrodes are applied to skin coated with conductive gel, and the signals are recorded on graph paper.

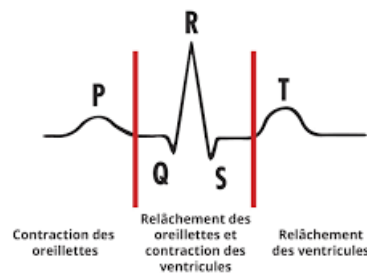
Wave P: represents the depolarization and contraction of the atria.

Intervals PR: corresponds to the conduction time of the electrical impulse from the atria to the ventricles.

Complex QRS: reflects the depolarization and contraction of the ventricles.

Segment ST: reflects the initial phase of ventricular repolarization.

Onde T: indicates repolarization and the return to electrical rest of the ventricles.



III Therapeutic applications of electricity

1. Defibrillation in the intensive care unit

Defibrillation (or cardioversion) involves delivering a brief electric current to the heart to correct certain rhythm disorders and restore a normal heartbeat.

Types of defibrillators:

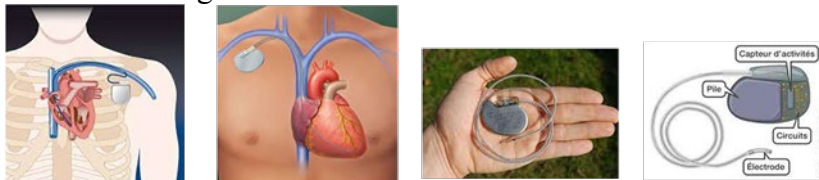


👉 **External Defibrillator:** a portable device analyzing the heart rhythm and delivering an electric shock if necessary.

👉 **Implantable Cardioverter Defibrillator:** Internal, autonomous device, implanted under the skin, connected to the heart. It automatically detects and corrects serious rhythm anomalies.

2) Biomedical application: Pacemaker

A pacemaker, or cardiac pacemaker, is an electronic device implanted to regulate the heart rhythm when it is too slow, too fast, or irregular. It consists of a casing (generator) containing a battery and circuits and leads that transmit electrical impulses to the heart. The implantation is generally performed through a small incision below the collarbone.



Functioning: The pacemaker monitors the electrical activity of the heart and delivers electrical impulses if necessary to maintain a sufficient and regular heart rhythm.

- Some pacemakers can adapt to the patient's exertion by accelerating the heart rate.

Implantation:

- The procedure is performed under local anesthesia and lasts about an hour.
- An incision is made under the clavicle, and the leads are inserted into a vein to guide them to the heart chambers, then connected to the casing.
- The casing is placed in a pocket under the skin.

Follow-up and precautions: Regular follow-ups are necessary to check the condition of the battery and the proper functioning of the device.

- Certain precautions must be taken to avoid interference, such as avoiding prolonged proximity to certain electronic devices or undergoing a manual search at airports.
- It is important to consult a doctor if symptoms such as dizziness, palpitations, or respiratory problems appear.

Types de pacemakers :

A. Conventional pacemakers: These devices are connected to the heart by one or more intracardiac leads. The generator, implanted under the skin, must be replaced on average every **8 to 10 years**, when the battery reaches the end of its life.

B. Leadless pacemakers (leadless, ex. Medtronic Micra): They are implanted directly into the right ventricle and do not require any leads. This technology is indicated for patients for whom the implantation of a conventional pacemaker is difficult or contraindicated (infection risk, lack of venous access, lead-related complications).

Operating principle of a pacemaker: The pacemaker is an electrical impulse generator based on an **RC** (resistor–capacitor) circuit.

- **Charging phase (switch position 1)** : The capacitor charges rapidly from the battery, through a low charging resistance.
- **Discharge phase (position 2)** : The capacitor discharges slowly through the equivalent resistance of the heart, which is of high value.
When the voltage across the capacitor reaches a **predefined threshold**, an electrical impulse is delivered to the myocardium, triggering cardiac contraction.

Illustrative numerical example

- Battery voltage: $U \approx 5\text{--}6\text{ V}$
- Battery capacity: 1–2 Ah (ampère-heure)
- Voltage applied to the heart: $U = 2\text{ V}$
- Equivalent cardiac resistance: $R = 2 \times 10^6\ \Omega$

Delivered current: $I = \frac{U}{R} = \frac{2}{2 \times 10^6} = 1\ \mu\text{A}$

Dissipated power: $P = U \cdot I = 2\ \mu\text{W}$

👉 These very low values explain the **long battery life** of pacemakers and their **energy safety** for cardiac tissues.

Difference between pacemaker and implantable cardioverter-defibrillator (ICD): The distinction is primarily based on the **therapeutic function** and the **intensity of the energy delivered**.

⚡ Implantable Cardioverter-Defibrillator (ICD)

- **Role:** to continuously monitor the heart rhythm and treat serious ventricular arrhythmias (ventricular tachycardia or ventricular fibrillation).
- **Action:** deliver a **high-energy electric shock** when the rhythm becomes chaotic.
- **Objective:** prevent sudden death and restore a rhythm compatible with life.
- **Analogy:** an internal defibrillator, always ready to intervene.

💓 Pacemaker (cardiac stimulator)

- **Role:** correct rhythms that are too slow (bradycardia).
- **Action:** emit low-amplitude, regular, and controlled electrical impulses.
- **Objective:** maintain a heart rhythm adapted to physiological needs.
- **Analogy:** a conductor setting the tempo for the heart.

3. Thermal ablation of lesions

It is a **medical technique** that involves **destroying abnormal tissues (lesions)** using **heat**.

Concretely:

- A **fine probe** (or needle) is inserted into the area to be treated.
- This probe heats up (using an electric current, microwaves, or a laser).
- The **heat selectively destroys diseased cells** without major surgery.

- 💡 This method is used, for example, to treat:
- Some **tumors** of the liver, kidneys, or lungs,
 - **Heart rhythm disorders** (radiofrequency ablation).

In summary: it is a minimally **invasive technique**, which **burns abnormal tissues** in a **precise and controlled manner**, while **sparing the surrounding healthy tissues**.

III- Dangers of electricity on the human body

Fundamental electrical quantities and electrical danger

The danger of electricity for the human body depends primarily on four electrical quantities: current intensity, voltage, resistance, and the power of the path traversed by the current.

- **Intensity I (ampere, A):** It is the flow of electrons colliding in the wires. The larger this flow, the higher the risk of electrocution.
- **Tension U (volt, V):** Tension (U) is expressed in volts (V). It is the force with which electrons are set in motion in electric wires. The higher it is, the greater the risk of electrocution.
- **Resistance R (ohm, Ω):** offered to the passage of electric current by any material expressed in ohms. The higher it is, the lower the risk of electrocution.
- **The dissipated electrical power is: $P = UI = RI^2$.** Electrical risk depends primarily on the intensity of the current passing through the body, but also on the applied voltage and the resistance of the path taken by the current (dry or wet skin, surface area and contact points, etc).

The orders of magnitude of body resistance vary significantly depending on the conditions, ranging from a few hundred ohms to several tens of kilo-ohms.

👉 Exposure to electricity can lead to dangerous effects on the human body.

Direct physical risks: Electric shocks: The passage of electric current through the body can cause violent muscle contractions, internal and external burns, and tissue damage.

Burns: Electricity can cause severe burns both externally and internally, particularly at the current's entry and exit points.

Effects on the cardiac system:

Heart rhythm disorders: Electric current can disrupt the natural functioning of the heart, causing potentially fatal arrhythmias.

Cardiac arrest: A sufficient electric shock can completely stop cardiac activity.

Neurological consequences:

- 🏠 **Brain injuries:** Electricity can damage the central nervous system, leading to permanent neurological disorders.
- 🏠 **Paralysis:** Nerves can be affected, causing temporary or permanent paralysis.

**Aggravating factors:**

Current intensity: The stronger the current, the more severe the damage is.

Exposure time: Prolonged contact significantly increases the risks.

Path of the current: Passing through the thorax or the brain is particularly dangerous.

Taking precautions (protective equipment, compliance with safety standards) is essential to avoid these potentially fatal dangers.

Summary table of the positions of Galvani and Volta.		
Aspect	Luigi Galvani (1737-1798)	Alessandro Volta (1745-1827)
Observation	Frog muscle contraction in contact with metals.	Same phenomenon but attributed to the metals themselves.
Interpretation	Existence of an "animal electricity" produced by living tissues.	Electricity comes from the contact of two different metals connected by a wet conductor.
Scientific contribution	Discovery of bioelectricity (nerve and muscle signals).	Invention of the voltaic pile (1800), the first source of continuous electricity.
Impact	Foundation of modern neurophysiology (ECG, EMG).	Beginning of electrochemistry and industrial electrical applications.
Vision	Living organisms are generators of electricity.	Metals are generators of electricity; living organisms are merely conductors.

Simple summary:

Property	Conductor	Insulator
Movement of charges	Free	Very limited
Inside electric field	Zero at equilibrium	May exist
Distribution of charges	On the surface	Remain inside
Examples	Copper, aluminum, silver	Glass, plastic, wood

5. Units:

Physical Quantity	Symbol	SI Unit	Expression
Current intensity	I	Ampere (A)	Coulomb per second (C/s)
Current density	j	Ampere per square meter ($A \cdot m^{-2}$)	$j = I/S$
Volume charge density	ρ	Coulomb per cubic meter ($C \cdot m^{-3}$)	$\rho = q/V$

Resistivity and electrical conductivity of some materials and biological media

Material	Resistivity ρ ($\Omega \cdot m$)	Conductivity γ ($S \cdot m^{-1}$)
Copper (Cu)	1.7×10^{-8}	5.8×10^7
Aluminium	2.8×10^{-8}	3.5×10^7
Human Body (wet fabric)	1 – 10	0.1 – 1
Pure water	10^6	10^{-6}

Illustrative diagram — Joule Effect

Element	Relation	Signification
Electrical power	$P = U \times I$	The power dissipated corresponds to the electrical energy consumed per unit of time.
Forms with the resistance	$P = R \times I^2$	The heat produced increases with the square of the current intensity and depends on the resistance of the conductor.
Total energy dissipated	$W = R \times I^2 \times t$	It is the total energy transformed into heat during the duration t of the current passage.
Observed effect	—	Heating of the conductor: temperature rise due to collisions of electrons with the atoms of the material.

Factors influencing resistance

Factor	Influence on R	Explication
Temperature	$\uparrow T \Rightarrow \uparrow R$ (for metals)	Collisions increase between electrons and atoms
Material	depends on the resistivity ρ	Copper: good conductor; Glass: Insulating
Dimensions	$R \propto l / S$	The longer and thinner the thread, the more resistant it will be

Summary diagram – Electrokinetics

Concept	Definition / Expression	Unit (SI)	Essential remarks
Electric current	Ordered movement of charges	A (ampère)	Conventional direction: $+\rightarrow -$
Intensity $I = \frac{dq}{dt}$	Charge passing through a cross-section per unit of time	A	Measured by an ammeter
Current density $\vec{j} = \rho \vec{v}$	Current per unit area	A/m ²	$I = \iint_S \vec{j} \cdot \vec{n} \, ds$
Ohm's Law (local) $\vec{j} = \gamma \vec{E}$	Current per unit area	—	γ : Material conductivity
Ohm's Law (global) $U = RI$	Voltage proportional to the current	V	$R = \rho \frac{l}{S}$
Electrical power $P = UI = RI^2$	Energy transferred per second	W (watts)	Joule's Law
Energy dissipated $W = RI^2t$	Heat released by the Joule effect	J (joule)	—
Resistors in series	$R_{eq} = R_1 + R_2 + \dots$	Ω	Same current, voltage shared

Concept	Definition / Expression	Unit (SI)	Essential remarks
Resistors in parallel	$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$	Ω	Same voltage, current shared
Real generator	$E = U + rI$	V	E : e.m.f. ; r : Internal resistance
Real receiver	$U = e + rI$	V	e: Counter-electromotive force
Node law	$\sum I_{incoming} = \sum I_{outgoing}$	—	Conservation of charge

IV. Biological applications

Application	Description	Connection with resistors
Biological tissues	The human body is a set of resistors in parallel (skin, blood, organs).	The overall resistance depends on the path of the current.
Medical Electrodes	Each electrode has a different contact resistance.	The series/parallel combination influences the current distribution.
Pacemaker	The internal components form resistive networks.	Optimization to limit heat (Joule effect).

Synthesis of fundamental laws and applications of electricity

Mesh law	$\sum U = 0$	—	Conservation de l'énergie
Wheatstone Bridge	$\frac{R_1}{R_2} = \frac{R_x}{R_v}$	—	Mesure de résistance inconnue
Electrical hazards	$U = RI, P = UI$	—	It depends on the resistance of the body

Comparative summary between resting potential and action potential

Criterion	Resting potential	Action potential
State of the cell	At rest	Excited/stimulated
Typical Value	~ -70 mV	From -70 mV to +30 mV (pico)
Duration	Stable	Very short (~1-2 ms)
Ion displacement	Mainly K ⁺	Na ⁺ and K ⁺ at high-speed
Function	Maintain the resting state	Propagate a nerve/muscle signal
Type	Stationary	Transient, "all or nothing"

Interpretation of the ECG

Element	Signification
Wave P	Depolarization of the earbuds
Space PR (or PQ)	Cardiac Conduction towards the ventricles
Complexe QRS	Ventricular Depolarization
Segment ST	End of repolarization
Wave T	Repolarization (relaxation) of the ventricles

In summary :

Caractarestic	DAI ⚡	Pacemaker ❤️
Principle role	Correcting dangerous heart rhythms	Regulating a pace that is too slow
Action type	Strong electric shock	Weak electrical impulses
Objectif	Saving life in the event of a stoppage	Maintain a steady pace
Usage	Serious Rhythm Disorders	Bradycardia (Slow Rhythm)

Exercise 1 : Calculate the equilibrium potential for K^+ , Na^+ , and Cl^- at $37^\circ C$, if intracellular and extracellular concentrations were equal : $[K^+]_i = 140 \text{ mM}$, $[K^+]_e = 5 \text{ mM}$ $[Na^+]_i = 15 \text{ mM}$, $[Na^+]_e = 145 \text{ mM}$ $[Cl^-]_i = 10 \text{ mM}$, $[Cl^-]_e = 110 \text{ mM}$.

The **equilibrium potentials** (or Nernst potentials) for each ion X.

We use **Nernst's formula** :

$$E_x = \frac{RT}{zF} \ln \frac{[X]_e}{[X]_i}$$

where:

- E_x is the equilibrium potential of l'ion X(in volts)
- $R=8.314 \text{ J.mol}^{-1}$.
- $T = 37^\circ=310\text{K}$
- $F = 96485 \text{ C/mol}$ (Faraday's constant)
- z = Valence of the ion
- $[X]_e$ et $[X]_i$ are the extracellular and intracellular concentrations.

At body temperature, the formula can be simplified by mV:

$$E_x \approx \frac{61}{z} \log_{10} \frac{[X]_e}{[X]_i} \text{ (in mV)}$$

- 1 For K^+ :** $[K^+]_i = 140$, $[K^+]_e = 5$, $z = +1$ $E_K \approx 61 \times (-1,448) \approx -88,3 \text{ mV}$.
- 2 For Na^+ :** $[Na^+]_i = 15$, $[Na^+]_e = 145$, $z = +1$ $E_{Na} \approx 61 \times 0,986 \approx +60,1 \text{ mV}$.
- 3 For Cl^- :** $[Cl^-]_i = 10$, $[Cl^-]_e = 110$, $z = -1$ $E_{Cl} \approx -61 \times 1,041 \approx -63,5 \text{ mV}$.
- ✓ Results :** $E_K \approx -88 \text{ mV}$, $E_{Na} \approx +60 \text{ mV}$, $E_{Cl} \approx -64 \text{ mV}$.

Exercise 2 : If the membrane potential of a cell is -70 mV and that the equilibrium potentials are $E_K = -90 \text{ mV}$ and $E_{Na} = +60 \text{ mV}$, determine:

1. The electrochemical force on K^+
2. The electrochemical force on Na^+
3. In which direction would these ions move spontaneously

1 Formula of the electrochemical force: $\Delta\mu_x = V_m - E_x$

- If $\Delta\mu_x < 0$ for a cation (+), the ion tends to **enter the cell**.
- If $\Delta\mu_x > 0$ for a cation, the ion tends to **come out**.
- For an anion (-), It's the opposite:
 - $\Delta\mu < 0 \Rightarrow$ The anion exists
 - $\Delta\mu > 0 \Rightarrow$ The anion enters

2 Calculation for K^+ $\Delta\mu_K = V_m - E_K = -70 + 90 = +20 \text{ mV}$

- It's a **cation**, so it's a positive sign $\rightarrow K^+$ will **come out of the cell**.

3 Calculation for Na^+ $\Delta\mu_{Na} = V_m - E_{Na} = -70 - 60 = -130 \text{ mV}$

- It is a **cation**, a negative sign $\rightarrow Na^+$ **will enter the cell**.

Exercise 3: A nerve cell has a resting potential of -70 mV . When stimulated, the potential reaches $+40 \text{ mV}$ before returning to rest.

Calculate: The amplitude of the action potential and The total potential change The energy required (assuming a capacity of $1 \text{ }\mu\text{F/cm}^2$)

1 Data

- Resting potential: $V_{\text{repos}} = -70 \text{ mV}$
- Maximum potential during the (PA): $V_{\text{PA}} = +40 \text{ mV}$
- Membrane capacity: $C = 1 \mu\text{F}/\text{cm}^2$

2 Magnitude of the action potential: The amplitude is the **difference between the maximum potential and the resting potential:**

$$\text{Amplitude} = V_{\text{PA}} - V_{\text{rest}} = 40 - (-70) = 110 \text{ mV}$$

3 Total Potential Change: The **total potential change** corresponds to the passage of -70 mV to +40 mV then back :

$$\Delta V_{\text{total}} = |V_{\text{PA}} - V_{\text{rest}}| + |V_{\text{rest}} - V_{\text{PA}}| = 110 + 110 = 220 \text{ mV}$$

4 Energy required to charge the membrane : The energy stored by a capacitor is given by :

$$E = \frac{1}{2} C (\Delta V)^2$$

- For a single depolarization (rest $\rightarrow$ +40 mV): $\Delta V = 110 \text{ mV} = 0,110 \text{ V}$

$$E = \frac{1}{2} \times 1 \times 10^{-6} \times (0,110)^2$$

$$E \approx 6,05 \times 10^{-9} \text{ J}/\text{cm}^2 = 6.05 \text{ nJ}/\text{cm}^2$$